Experimental Evaluation of a Financial Education Program in Elementary and Middle School Grades[*]

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Abstract
This paper investigates whether providing financial education in elementary and middle school grades improves students' financial proficiency and actual behavior. We use a cluster randomized controlled trial to evaluate a pilot program across 201 Brazilian municipal schools, 101 of which received the intervention in 2015. The findings show positive impacts on financial proficiency (0.07 of a standard deviation), mainly among middle school students, a significant shift in hypothetical attitudes toward saving and consumption, especially among fifth graders, and suggestive evidence of improvements in short-term behavioral outcomes. However, the analysis indicates that the program did not impact students' school achievement up to three years after the intervention, which suggests that its effects were not strong enough to shift students' human capital decisions.
JEL Codes: D14, I21, I25, O12, O15

1 Introduction

Financial literacy is at the center of the education debate. As individuals are increasingly exposed to financial decisions at a young age, the sooner they learn to incorporate the costs of their choices into their decision-making, the better. The Organization for Economic Cooperation and Development (OECD), for instance, recommends that financial education start at school, so that people are educated about financial matters as early as possible in their lives (OECD, 2005, Good Practice 9).[1] In fact, financial education interventions have been implemented in schools around the world, aiming to develop financial skills that could be conducive to better decision-making throughout individuals' life-cycle (Bruhn et al., 2016; Berry et al., 2018; Luhrmann et al., 2018; Gill and Bhattacharya, 2019; Batty et al., 2020; De Beckker et al., 2021; Frisancho, 2023a; Bover et al., 2026). Vivian Amorim..

[*]Acknowledgements: We would like to thank Cláudia Forte, Alzira Silva, Thiago Nascimento, and Cláudia Donega from the Brazilian Association of Financial Education (AEF-Brasil), José Alexandre Vasco from the Securities and Exchange Commission of Brazil (CVM), and the staff from Municipal Secretariats of Education from Manaus and Joinville. We thank the seminar participants from the Sao Paulo School of Economics at Getulio Vargas Foundation, Insper Institute of Education and Research, the 39th Brazilian Econometrics Society Meeting (SBE), and the 3ie-IFPRI Seminar Series. We also thank Arianna Legovini, Bilal Zia, Miriam Bruhn, Florence Kondylis, and an anonymous referee for their comments and suggestions. Finally, we thank DIME i2i, DEC Research Support Budget, and the Brazil Country Office for funding support. The reproducibility package is available on the GitHub repository financial-education-brazil. The authors made extensive use of generative artificial intelligence tools (Anthropic's Claude) in preparing this paper, including assistance with the analysis code, tables, figures, and editing of the text. All AI-assisted output was carefully reviewed and verified by the authors, who take full responsibility for the content of the paper.

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[1]In this study, we use financial education and financial literacy interchangeably.

Most of the existing evidence on the impact of financial education, however, comes from programs aimed at adults, and early results were mixed at best: interventions explained a negligible share of the variance in financial behaviors (Fernandes et al., 2014) and modestly improved savings behavior but not loan repayment (Miller et al., 2015). On the other hand, a recent meta-analysis based on 76 randomized controlled trials shows that financial education interventions have a significant impact on financial knowledge (about 0.2 of a standard deviation) and on financial behaviors such as saving, borrowing, and budgeting (about 0.1 of a standard deviation) (Kaiser et al., 2022).[2]

The ultimate goal of financial education interventions is to change individuals' behavior. One can think of financial education affecting behavior through at least two channels. The first is financial knowledge itself: financially literate individuals are less likely to make costly financial mistakes, such as high-cost borrowing, and are more likely to take advantage of economic opportunities, such as retirement planning and participation in financial markets (Lusardi and Mitchell, 2014). In incentivized convex time budget experiments in Germany, a financial education program made high school students' intertemporal choices more time-consistent, though not more patient (Luhrmann et al., 2018). An observational study with college students shows that with higher pre-existing financial knowledge made seemingly more patient choices, a pattern attributed to knowledge-driven arbitrage rather than to deeper patience (Oberrauch and Kaiser, 2022). Nonetheless, whether gains in financial knowledge drive changes in behavior is rarely tested directly.

The second channel is the shaping of attributes such as self-control and the ability to make forward-looking decisions, particularly in programs aimed at children. Such noncognitive attributes can shape schooling decisions and labor market prospects (Heckman et al., 2006; Kautz et al., 2014; Acosta and Muller, 2018). In particular, more impatient children and adolescents are less likely to invest in human capital and more likely to consume alcohol, develop obesity, and become pregnant during adolescence, while those with lower self-control are more likely to commit crimes later in life (Castillo et al., 2011; Sutter et al., 2013; Golsteyn et al., 2014; Moffitt et al., 2011). These attributes are also malleable in childhood: a short classroom program, framed as financial literacy, that had young Turkish students imagine their future selves raised their patience in incentivized choices, an effect still detectable almost three years later, and lowered the incidence of behavioral problems at school (Alan and Ertac, 2018); a subsequent arm of the same experiment, targeting grit, increased students' perseverance and school performance (Alan et al., 2019), as did a nationwide grit curriculum in North Macedonia (Santos et al., 2021). Programs aimed at developing such attributes should prioritize early ages as they are easier to change and consequently can be more impactful and cost-effective (Cunha and Heckman, 2007).

[2]The meta-analysis covers more than 160,000 individuals; about three quarters of its treatment-effect estimates come from samples of adults, one fifth from youths aged 14 to 25, and fewer than one tenth from children under 14.

There is increasing evidence that financial education experiments with high school students are effective in increasing financial literacy, with effects ranging from 0.14 to 0.24 of a standard deviation (Bover et al., 2026; Frisancho, 2023a; Bruhn et al., 2016).[3] The available literature also points to changes in behavior: treated students save and budget more in a large experiment in Brazilian high schools (Bruhn et al., 2016), and these effects can be long-lasting; nine years after the program, treated students used less expensive credit, defaulted less, and were more likely to own a formal microenterprise (Bruhn et al., 2025). In Peru, seven years after school-based lessons, treated individuals shifted their borrowing toward productive small-business loans on slightly better terms, suggesting credit-financed entrepreneurial activity, although formal business creation is unchanged (Frisancho et al., 2025). In Spain, treated students make more patient choices in incentivized tasks months after the course ends (Bover et al., 2026).

The existing evidence on the impacts of financial literacy interventions in elementary and middle school grades is still limited, particularly for developing countries. In an experiment in Belgium, De Beckker et al. (2021) find that a four-hour online course increased the financial knowledge of eighth and ninth graders by 0.46 of a standard deviation,[4] though without a corresponding change in their consumer choices. Batty et al. (2020) find an increase of about 0.25 of a standard deviation in the financial knowledge of fourth graders in an experiment with an experiential "classroom economy" program across 20 schools in the United States. Berry et al. (2018), on the other hand, do not find impacts on financial literacy in an experiment across 135 Ghanaian elementary and junior high schools, with the programs shifting where children saved rather than raising their total savings.

In this paper, we use a cluster-randomized controlled trial to investigate the impact of a financial education pilot implemented in 2015 in elementary and middle school grades in two Brazilian municipalities (Manaus and Joinville). We randomly assigned 101 municipal schools to receive the program, and 100 to a control group. Rather than teaching financial facts alone, the intervention aimed to build the attitudes and everyday competencies behind sound financial choices: telling wants from needs, planning and consuming consciously, saving, and weighing the present against the future consequences of one's decisions. The materials anchor these competencies in everyday situations, with students reading real utility bills, comparing cash and installment prices, budgeting school events, and, in the later grades, making decisions inside stories and games that expose common psychological traps, such as ostentation and immediacy. In short, the program sought to help young students become more conscious of the trade-offs embedded in their daily decisions and more forward-looking. The evaluation focused on grades three, five, seven, and nine for reasons discussed later.

We find positive effects on students' financial knowledge of 0.07 of a standard deviation, on

[3] In a quasi-experiment, Gill and Bhattacharya (2019) find evidence that twelfth graders taught financial concepts in U.S. economics classes gained about 13 percentage points on a financial literacy test.

[4] Financial knowledge was measured right after the end of the course.

average, driven by the middle school grades (0.1-0.15 SD). These magnitudes are in line with the evidence of financial education programs targeting children under age 14 (Kaiser et al., 2022). We also find evidence that the program made students' reported attitudes toward consumption and saving more prudent, particularly fifth graders. Futhermore, for the same age cohort, we find suggestive evidence of changes in some contemporary behavioral outcomes, such as talking about money with parents and friends, indicating linkages to short-term behavioral changes. However, we do not detect robust impacts on contemporary learning outcomes or on medium-term human capital investments. We use a mediation causal analysis to investigate the potential causal link between gains in financial knowledge and changes in behavioral outcomes. Our results indicate a weak link between increase in formal knowledge and changes in behavior, a result consistent with De Beckker et al. (2021).

Our paper has four main contributions. First, we document the positive impacts on financial proficiency, attitudes, and behavioral outcomes of a large-scale pilot intervention implemented in elementary and middle school grades. Second, we show that financial education programs can potentially affect different skills across students' life cycles. Third, we test the overlooked hypothesis that increasing financial proficiency is a necessary condition to change individuals' behavior. Finally, we use student-level data from the Brazilian national standardized exams to assess the program's impacts on students' contemporary and medium-term human capital.

Apart from this introduction, this paper is organized as follows. Section 2 describes the pilot program and its implementation. Section 3 describes the data collection. Section 4 introduces the empirical strategy, whereas Section 5 discusses the results. We then conclude.

2 The Pilot

In 2010, the Brazilian Federal Government established the National Strategy for Financial Education (ENEF) to promote financial education and empower individuals to make autonomous financial decisions.[5] In 2015, the Brazilian Association of Financial Education (AEF-Brasil),[6] ENEF's implementing partner, developed a financial education pilot targeting elementary and middle school students, in which the municipalities of Joinville and Manaus volunteered to participate. The intervention was delivered throughout the 2015 school year in a sample of public elementary and middle schools (grades 1–9) managed by the respective municipal governments. The initiative followed an earlier financial education program piloted under the same strategy across 892 public high schools across six Brazilian states, which yielded significant improvements in students' financial knowledge, intention to save, and financial autonomy, as well as greater participation in their household finances (Bruhn et al., 2016).

The two pilot municipalities have distinct socioeconomic contexts. Manaus, the capital of Amazonas, is in the less affluent North, while Joinville, in Santa Catarina, is in the relatively affluent South. At the time of the pilot, Joinville exhibited higher values on development indicators, including GDP per capita (USD 23,499 vs. USD 16,218, in 2015 PPP terms), the Municipal Human Development Index (0.809 vs. 0.737), and school attendance among 6- to 14-year-olds (97% vs. 94%).[7] Together, the two settings span two of the major socioeconomic profiles found in the Brazilian education system, broadening the external validity of our findings within the country while not exhausting its full diversity.

2.1 Teaching Materials

ENEF's teaching materials comprise one student textbook and one teacher's book for each of grades 1–9, of which the pilot evaluates a subset, described below.[8] These materials follow the OECD's definition of financial education as the process by which individuals improve their understanding of financial concepts and risks and develop the skills and confidence to make

[5] ENEF stands for *Estratégia Nacional de Educação Financeira* and was created by a Federal Decree – 7.397 of December 22, 2010. It brings together the Ministry of Education, the Central Bank of Brazil, the Securities and Exchange Commission of Brazil (CVM), the Ministry of Finance, the Superintendence of Private Insurance (SUSEP), and the Superintendence of Complementary Pension (PREVIC).

[6] AEF-Brasil stands for *Associação de Educação Financeira do Brasil*.

[7] GDP per capita from the 2015 Municipal Accounts of the Brazilian Institute of Geography and Statistics (IBGE): R$ 47,233 for Joinville and R$ 32,598 for Manaus, converted to international dollars using the 2015 PPP conversion factor for Brazil of R$ 2.01 per international dollar (World Bank). Municipal Human Development Index and school attendance rates from the 2010 Demographic Census, the latest available at the time of the pilot, disclosed by the IBGE.

[8] Materials are publicly available on the CVM's investor education portal at https://gmw.investidor.gov.br/ensino-fundamental/. They were published in 2014 by the National Committee of Financial Education (CONEF) and validated by an inter-institutional pedagogical panel.

informed financial decisions (OECD, 2005).[9] This process operates through three means: (1) information (facts and data enabling sound financial choices and an understanding of their consequences), (2) instruction (the competencies needed to select and apply that information), and (3) objective advice on specific financial products. The teacher's books state that basic-education materials deliberately address only the first two, since advice on financial products is aimed at adults, and that greater emphasis should be placed on instruction than on information given the students' age.

The content is organized along two dimensions: a spatial dimension that moves from the individual to local, regional, national, and global levels, emphasizing how individual choices reverberate at larger scales; and a temporal dimension that connects past, present, and future decisions, placing the intertemporal nature of financial choice at the center. These dimensions are operationalized through four pedagogical modules, each targeting a grade range: a project-based approach for grades 1–4, a choose-your-own-adventure narrative for grades 5–6, a role-playing simulation game for grades 7–8, and a news-magazine format for grade 9.[10] Table 1 shows examples for the four pilot-grade textbooks.

The content was not designed to be delivered as a stand-alone subject. Instead, it was envisaged to be integrated across subjects of the existing curriculum and to be taught by teachers of any specialty, drawing on what the teacher's books describe as the teacher's role as a citizen rather than as a subject specialist. This integration is built into the material itself: the margins of the teacher's books contain icons flagging, for example, connections to specific curricular areas (Portuguese, Mathematics, Environmental Education); the introduction of formal financial concepts (e.g., compound interest, household budget); or psychological biases (e.g., loss aversion, immediacy). Because each grade range receives a distinct combination of content and pedagogical format, our estimates capture the joint effect of these elements.

The teacher's books also present the program as complementary to students' learning in Portuguese and Mathematics, with reading-comprehension and mathematical-reasoning activities built into the financial content across all grades; the books for grades 1–4 go further, explicitly committing the program's pedagogical project to improving students' results in these two subjects. Since these are the subjects assessed in the national standardized exams, our analysis of school achievement in section 5 directly tests this stated ambition.

[9] The curriculum has 10 learning competencies: debating rights and duties (C01); participating in socially and environmentally responsible financial decisions (C02); distinguishing wants from needs of consumption and saving within the family's life project (C03); reading and interpreting simple texts in the financial education domain (C04); critically reading advertising texts (C05); participating in financial decisions considering real needs (C06); acting as a multiplier (C07); elaborating a basic financial plan with assistance (C08); and caring for oneself, nature, and common goods, considering both the immediate (C09) and future (C10) consequences of present actions.

[10] Each pilot-grade teacher's book describes its module's pedagogical approach in Part I, Sections 4 and 5.

Table 1: Financial education materials by pedagogical module

Module (grade range)	Pedagogical format	Core content
Project-based (grades 1–4)	Four projects per grade built around objects of everyday life; in third grade: a ball, the household's bills, the school, and a book fair.	Where prices come from; needs vs. wants; household income and expenses; care for shared public and private spaces; short-term planning and estimation.
Choose-your-own-adventure (grades 5–6)	Three branching stories centered on sustainable consumption (the "five R's": rethink, refuse, reduce, reuse, and recycle), each with financially better and worse routes, plus a closing group project with a real budget.	Conscious consumption; value vs. price; cash vs. installment purchases; saving; risk; first psychological traps (e.g., ostentation, immediacy).
Role-playing (grades 7–8)	A six-session game in which student groups play community organizations, negotiate proposals, and manage a running budget; one financial concept is added per session.	Personal budgeting; investment and the risk–return trade-off; financing, borrowing, and interest; insurance and patrimony protection; further traps (e.g., framing, sunk costs).
News-magazine (grade 9)	A print magazine simulating a website, with reports, interviews, columns, a three-episode web-series, and a forum, closed by hands-on group activities.	Household budgeting and expense categories; credit cards and compound interest; consumer traps; consumer-protection law; public budget and taxes.

Notes: The table summarizes the four pedagogical modules of the program (covering grades 1–9). Each row describes one module and their respective grades.
Source: Student textbooks and teacher's books for grades 1–9, available at https://gmw.investidor.gov.br/ensino-fundamental/.

2.2 Sampling and Randomization

The sampling frame comprised 374 municipal schools offering elementary or middle school grades: 72 in Joinville and 302 in Manaus.[11] To account for differences in infrastructure and socioeconomic context across the two municipalities, we stratified by municipality and by school type: schools offering only elementary grades (grades 1–5), schools offering only middle school grades (grades 6–9), and schools offering all elementary and middle school grades (grades 1–9), giving six strata in total (Figure 1).

The 72 schools in the Joinville frame entered the study. In Manaus, the budget did not allow us to survey all 302 schools, so we drew 129 of them. All 65 schools offering grades 1–9 were included. In the two remaining strata, we ranked schools by their 2013 IDEB, a school-level index that combines student performance in reading and mathematics with grade promotion rates,[12] and drew the same number of schools at random from each quartile of that ranking: 9 schools per quartile among schools offering only elementary grades, totaling 36, and 7 schools

[11]The frame is documented in the pilot's implementation report, prepared by the World Bank in August 2016. Relative to the 2015 School Census, which records 369 municipal schools offering elementary and middle school education in Manaus and 83 in Joinville, the frame leaves out 67 schools in Manaus and 11 in Joinville. In Manaus, the majority of the schools out of the sample frame are in riverside communities that are very challenging to access for data collection. In Joinville, the 11 schools left out are all small rural schools.

[12]IDEB stands for *Índice de Desenvolvimento da Educação Básica*. A school's IDEB is measured in grade 5 for the elementary grades and in grade 9 for the middle school grades.

per quartile among schools offering only middle school grades, totaling 28.[13] Within each of the six strata, we then randomly assigned roughly half of the schools to treatment and half to control, independently of the IDEB ranking used to draw the sample, yielding 201 study schools, 101 treated and 100 control.

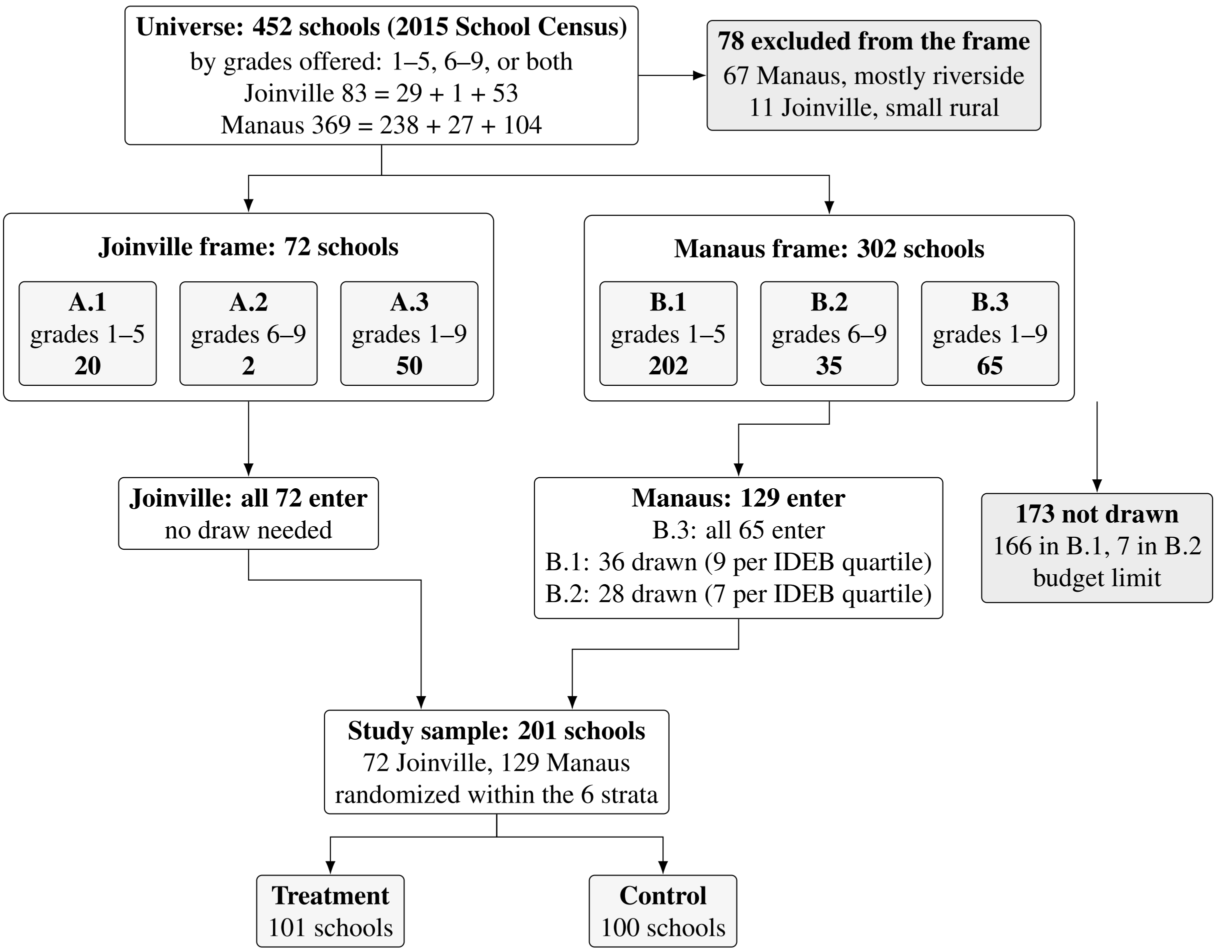


Figure 1: From the school universe to the randomized sample

Notes: The figure tracks the schools from the 2015 School Census universe to the randomized sample. School counts in the universe follow the census enrollment flags for each cycle (grades 1–5 and 6–9). The six strata are defined by municipality and school type: A.1–A.3 in Joinville and B.1–B.3 in Manaus, each in the order grades 1–5 only, grades 6–9 only, and grades 1–9. The strata classify schools by the grades targeted by the pilot: a school serves the elementary cycle if it offers grade 3, which reassigns nine schools offering grade 5 but not grade 3 (one in Joinville, eight in Manaus) to the 6–9 strata. In Manaus, the majority of the schools out of the sampling frame are in riverside communities that are very challenging to access for data collection; the 11 schools left out in Joinville are all small rural schools. The 72 schools in the Joinville frame entered the study, as did all 65 Manaus schools offering grades 1–9 (stratum B.3). In strata B.1 and B.2, the budget did not allow us to survey every school, so we ranked schools by their 2013 IDEB, a school-level index that combines student performance in reading and mathematics with grade promotion rates, and drew the same number of schools at random from each quartile of that ranking. Within each of the six strata, we then randomly assigned roughly half of the schools to treatment and half to control, independently of the IDEB ranking; stratum B.3 holds an odd number of schools, hence 33 treated and 32 control.
Source: Sampling frame from the pilot's implementation report, prepared by the World Bank in August 2016; school counts for the comparison with the census from the 2015 School Census (INEP).

Budgetary constraints also shaped other aspects of the design. First, the pilot could not include all elementary and middle grades. In agreement with AEF-Brasil, we selected grades 3, 5, 7, and 9 to generate evidence on each of the four pedagogical modules shown in Table 1. Second, we did not conduct a baseline student survey and instead relied on rich school-level administrative

[13] Taking the same number from every quartile ensures that the sample spans the full range of school quality in the municipality rather than concentrating on schools of average performance.

data to perform pre-treatment balance tests (section 3). Third, within each school, the program was not delivered to every class of the pilot grades: at the beginning of the 2015 school year, each school principal, in both treatment and control schools, selected a single class at each pilot grade to be included in the survey sample. Across the 201 pilot schools, the median number of classes per pilot grade was three,[14] so principals typically chose one of about three classes per grade.

This within-school selection of classes could be a potential source of bias. Because randomization preceded principals' class selection, principals in treated schools knew their school's assignment and may have selected the classes to receive the program strategically, for example, choosing their strongest classes, a possibility their control-school counterparts had no reason to mirror. As we show in section 3, however, students' socioeconomic characteristics, teachers characteristics, and pre-treatment student-level performance in standardized test scores are balanced across treatment arms. The only exception is ninth graders' reading test scores obtained in grade 5. Students in the control group display higher pre-treatment reading scores. If higher reading scores in grade 5 correlates (predicts) financial proficiency, this imbalance would bias our estimates downward. As a robustness check, we show treatment effect estimates controlling for baseline scores in standardized tests whenever the information is available.

2.3 Program Implementation

ENEF trained local education officials on how to incorporate the financial education textbooks into the 2015 school-year curriculum, which ran from February to early December.[15] Schools had the autonomy to decide how many teachers would be in charge of delivering the program's content for each of the pilot grades, the subject each of those teachers taught, and when exactly in the academic year to introduce the textbook into their classes.

The pilot data show that in 78.9 percent of treated classes only one teacher was in charge of the financial education content, whereas in 14.4 percent of them, two teachers were in charge, and in the remaining 6.7 percent three teachers shared this responsibility.[16] Among the classes with more than one teacher in charge of the pilot content, 85.7 percent are seventh- and ninth-grade classes, consistent with the fact that these grades have multiple teachers.

In the early grades, the content was delivered by generalist teachers: this is the case for every teacher in charge of it in treated third-grade classes and for 75 percent of those in charge in the

[14] 2015 School Census.

[15] The training reached teaching supervisors in Joinville in February 2015 and supervisors of the regional education boards in Manaus at the end of March; these supervisors, in turn, trained the teachers in the pilot schools.

[16] Not all teachers of the classes included in the pilot answered the teacher questionnaire: 4.8 percent of treated classes and 6.1 percent of control classes have no teacher information, a difference that is not statistically significant (p-value of 0.62 from a regression with strata dummies and standard errors clustered at the school level). The shares in the text are computed among treated classes with teacher information.

fifth grade, with the rest being subject teachers, most often of mathematics.[17] In the middle-school grades, where each class has several subject teachers, responsibility was spread across the full range of specialties, most often mathematics (31.3 percent) and Portuguese (21.6 percent) but reaching science, geography, history, physical education, and others.[18]

The teacher questionnaires suggest that the training did not reach every teacher in treated schools. Table A.1 shows that 23.6 percent of teachers in the treatment group reported not receiving any training on how to use the financial education textbook. The pilot data show that 44 of the 101 treated schools had at least one teacher in this situation. Despite that, 95.5 percent of teachers in treated schools reported delivering at least one financial education class, and a lower percentage reported using the students textbook (90.3 percent).[19]

The survey with teachers also suggest that the delivery started late. About two fifths (41.4 percent) of teachers reported delivering the content only in the second semester, against 23.3 percent who delivered it in the first semester and 29.8 percent who spread it over the whole year. The late start is more pronounced in middle school, where 49 percent of teachers delivered the content only in the second semester (Table A.1). This is reflected in coverage: one fourth of treated teachers reported covering more than 80 percent of the textbook, 31.6 percent in the elementary grades and 19.7 percent in middle school. Based on these implementation issues, the program should then be taken as a low-intensity intervention.

The pilot questionnaires also point to some contamination of the control group. On the teacher side, 2.3 percent of control-school teachers reported receiving the training, 3.7 percent reported receiving the teacher's book, and an even higher percentage, 11.3 percent, reported delivering at least one financial education class (Table A.1). On the student side, 16.3 percent of control students reported having had financial education classes, 20.4 percent reported receiving the textbook at some point in the year, and 7.7 percent reported that their teacher used it; the two textbook reports carry more noise, as we cannot rule out that some students had other teaching materials in mind.

Nonetheless, the teacher and the student answers on exposure to financial education barely coincide. There are 28 control classes whose teacher reported having taught the content and 21 classes where half or more of the students reported having had them, but only one class has

[17] Shares computed with data from the pilot teacher questionnaire considering the teachers in charge of the financial education content in treated elementary-school classes. In the fifth grade, 75 percent of them are generalist teachers, 8.7 percent teach mathematics, 6.5 percent physical education, 5.4 percent Portuguese, 2.2 percent science, 1.1 percent English, and 1.1 percent art.

[18] Shares computed with data from the pilot teacher questionnaire considering the teachers in charge of the financial education content in treated middle-school classes. Science accounts for 8.7 percent, geography for 8.7 percent, history for 6.7 percent, physical education for 6.7 percent, English for 5.3 percent, and art for 3.8 percent, with other subjects accounting for the remaining 7.2 percent.

[19] The questionnaire asked how many financial education classes the teacher had taught, with answer categories running from one to eight or more, so Table A.1 reports a share for each category. The 95.5 percent is the sum of those shares, and the remaining 4.5 percent are the teachers who left the question blank.

both.[20] One possibility is that these students were exposed to financial education concepts by teachers who were not part of the pilot, and whose answers we therefore do not observe; another is that they had other content in mind when answering. This imperfect two-sided compliance attenuates the treatment effect estimates and consequently increases the probability of type II error.

[20]Counted over the 210 control classes that have teacher information, in grades 5, 7 and 9, the ones that answered the socioeconomic questionnaire. Control students report having had classes at about the same rate whether or not their teacher reports teaching them, 22.7 against 21.5 percent, while in treated classes 86.9 percent of students report having had them.

3 Data and Measurement

To evaluate the impact of the pilot, we use a combination of survey and administrative data to measure a broad set of outcomes over different time horizons. Survey data collected at the end of the 2015 school year capture short-term effects on students' financial proficiency, hypothetical attitudes toward consumption and saving, and self-reported behaviors. The survey data were collected right after the program implementation finished, in line with other financial education programs targeting elementary and middle school students (De Beckker et al., 2021; Batty et al., 2020; Berry et al., 2018). Administrative records provide short- and medium-term human capital outcomes which we interpret as part of behavior changes the program intended to impact. These outcomes include grade progression, standardized test scores, and dropout rates, observed up to three years after the intervention.

3.1 Survey Data

The Center for Public Policy and Education Evaluation (CAEd), a center at the Federal University of Juiz de Fora specialized in standardized assessments and large-scale data collection, was responsible for implementing the survey, with students and teachers participating in the pilot. Data collection occurred during the final school days of the 2015 academic year, when CAEd distributed printed questionnaires directly in classrooms. Despite being conducted at the end of the school year, a period when some students may already be on vacation or have dropped out, student participation rates remained high, averaging approximately 80%. Furthermore, as shown in Table A.4, there is no significant difference between participation rates of treatment and control groups across all grades.

Students completed three paper-based questionnaires during a two-hour session: (i) a financial literacy test, (ii) a questionnaire on hypothetical consumption and savings attitudes and self-reported behaviors, and (iii) a socioeconomic questionnaire. The content of each instrument was tailored to the students' grade level and cognitive development. Fifth-, seventh-, and ninth-graders answered all three questionnaires in school. Third graders answered only the financial literacy test, which was read aloud by trained enumerators, given that students at this age are still developing reading and writing skills; the attitudes-and-behaviors questionnaire was not administered to this grade, and the socioeconomic questionnaire was sent to the students' homes for their legal guardians to fill out and return the following day.[21]

The financial literacy test evaluated students' understanding of concepts introduced in the financial education textbooks through multiple-choice questions aligned with grade-appropriate

[21] Schools prepared the legal guardians by sending them a note before the test day. Table A.4 shows that an average of 65% of parents/guardians returned the questionnaire, with no statistically significant difference between treated and control students.

skills (Table A.2). The test includes items that assess students' ability to interpret everyday financial information, such as utility bills or spending records, and to identify the purpose of documents like receipts, checks, and invoices. It also covers savings, expenses, consumption, waste, risk, return, financial planning, and investment. Additional questions assess students' ability to estimate values for simple financial projects, distinguish between paid and unpaid work, interpret financial graphs and tables, and extract relevant financial information from media texts. Not all skills are assessed in all grades; the test content is adapted to the cognitive level of each group, with some skills appearing only in earlier or later grades. CAEd used student responses to construct a financial proficiency index using Item Response Theory (IRT), which accounts for the difficulty of each question, generates a standardized score comparable across grades, and allows us to assess the impact of the intervention in terms of standard deviations.

The second questionnaire measures students' hypothetical attitudes toward consumption and saving and collects a set of self-reported behaviors. The attitudinal part consists of fifteen statements, seven on consumption and eight on saving, each printed with a four-point agreement scale running from "Totally agree" to "Totally disagree" (Table A.3). The statements are worded in both directions: agreeing with "I plan before spending my money" indicates a prudent attitude, while agreeing with "I see no problem in owing money" indicates the opposite. We reverse the scale of the positively worded statements so that in every item the highest value corresponds to the most prudent response. We summarize each block by its first principal component, standardized to mean zero and unit variance, and refer to the two resulting measures as the consumption and savings indices, an approach similar to Bruhn et al. (2016). Since this second questionnaire was not administered in third-grade, the consumption and savings indices, like the self-reported behaviors below, are available only for grades 5, 7, and 9. These are non-incentivized, self-reported measures of what students say about money.

The second questionnaire also collects self-reported behaviors: whether the student keeps a piggy bank, receives an allowance, holds a bank account, a debit card or a credit card, and whether they discuss money with their parents or with their friends. We analyze these separately, combining only the bank account, debit card, and credit card items into a single indicator of access to financial services.

The last questionnaire asks about students' socioeconomic background, such as the educational attainment of their mothers, and whether they benefit from the national conditional cash transfer program, *Bolsa Família*. As expected, Table A.4 shows no significant differences between the socioeconomic characteristics of treated and control students.

The teachers' questionnaire investigates their wage level, experience, and the incorporation of financial education textbooks in the curriculum, among other questions. Table A.5 shows that there are no significant differences between teachers in treatment and control groups.

Overall, our analysis uses eight dependent variables from the survey data collected in the field.

Three are standardized indices: (i) financial proficiency, computed by CAEd from the financial literacy test using Item Response Theory, and the (ii) consumption and (iii) savings attitude indices described above. The remaining five are self-reported behaviors: (iv) whether the student talks to their parents about financial subjects, (v) whether they talk to their friends about the same subjects, (vi) whether they keep a piggy bank, (vii) whether they have access to financial services, and (viii) whether they receive an allowance from their parents.

3.2 Administrative Data

Since 1995, every public and private K–12 school has reported to the annual School Census, implemented by the National Institute of Educational Studies and Research (INEP), a research agency under the Brazilian Ministry of Education. The Census records school infrastructure and students' outcomes such as age-grade distortion, grade promotion, retention, and dropout. Since 2005, INEP has also administered *Prova Brasil*, a standardized assessment of reading and math proficiency given every two years to all students in the fifth and ninth grades of public schools.[22] Finally, since 1998 INEP has run the *Exame Nacional do Ensino Médio* (*ENEM*), a national exam taken at the end of high school. Unlike the other two, *ENEM* is voluntary: students register on their own, and since 2009 most universities have used it for admission, which raised the number of registrations from about 157,000 in 1998 to 7.7 million in 2015.[23]

Because of the limited budget and the richness of these public records, we did not collect a baseline student survey. Instead, we used the administrative data at the school level to check pre-treatment balance between treatment and control schools: Table A.6 shows no significant differences in average proficiency in reading and math, grade promotion, dropout, retention, or age-grade distortion. Nonetheless, school-level balance has a limitation since not every student of a pilot school took part in the intervention: as explained in section 2, principals selected the participating class at each grade, and treatment- and control-school principals could in principle have chosen classes on different, unobserved criteria.

To compare treated and control students rather than schools, in 2019 we asked INEP to merge the pilot roster with its student-level administrative records.[24] Table A.7 and Table A.8 show no significant differences in the share of treated and control students that INEP located in its administrative datasets. The merge serves two purposes. First, it lets us check balance at the student level. For the seventh- and ninth-grade cohorts, we recover their grade 5 scores in *Prova Brasil*. Table A.8 shows that seventh-grade students are balanced on this pre-treatment score,

[22] Schools with at least 20 students enrolled in the evaluated grade. Students take the test at the end of the school year, between October and November.

[23] INEP's *ENEM* historical series records 157,221 registrations in the first edition (1998) and 7,746,057 confirmed registrations in 2015.

[24] Because researchers cannot access identified student-level microdata, the merge itself was performed by INEP, using students' names, grades, and schools' identification numbers. We then ran all balance tests and regressions on the merged data ourselves inside INEP's secure facility (the cold room).

whereas ninth-grade control students have significantly higher pre-treatment reading scores. If higher reading correlates with financial proficiency, this imbalance would bias our estimates downward. The third- and fifth-grade cohorts have no such baseline, because *Prova Brasil* is first taken in the fifth grade.

Furthermore, the merge enriches the analysis with outcomes that exist only in the administrative records. Table 2 lists them by cohort: post-treatment proficiency in reading and math in *Prova Brasil*, retention and dropout, observed annually from 2015 to 2018 in the School Census; and, for the ninth-grade cohort, high-school completion, *ENEM* participation, and *ENEM* proficiency scores.[25]

Table 2: Administrative outcomes recovered for pilot students, by 2015 cohort

	Pre-treatment	Post-treatment				
Cohort	Proficiency (*Prova Brasil*)	Proficiency (*Prova Brasil*)	Retention	Dropout (School Census)	HS completion	Participation & proficiency (*ENEM*)
third graders in 2015	—	while in fifth grade in 2017	2015–2018	2015–2018	—	—
fifth graders in 2015	—	while in fifth grade in 2015	2015–2018	2015–2018	—	—
seventh graders in 2015	while in fifth grade in 2013	while in ninth grade in 2017	2015–2018	2015–2018	—	—
ninth graders in 2015	while in fifth grade in 2011	while in ninth grade in 2015	2015–2018	2015–2018	2018	2018

Notes: Each row is a pilot cohort, defined by the grade the student attended in 2015, and each cell reports the year in which the outcome is observed for that cohort through the student-level merge with INEP's records. Dashes mark outcomes that do not exist for the cohort.

Prova Brasil is administered only in the fifth and in the ninth grades, so each cohort is tested when it reaches one of these grades. After the pilot, the fifth- and ninth-grade cohorts were tested in 2015, in the pilot year itself, and the third- and seventh-grade cohorts in 2017, when they reached the fifth and the ninth grade, respectively. Before the pilot, only the seventh- and ninth-grade cohorts had been tested, while in the fifth grade in 2013 and 2011, respectively. The years in the table are those of students who progressed on schedule; a student who repeated a grade reaches the fifth or the ninth grade, and hence the test, in a later year. The pre-treatment proficiency score is the outcome we use to assess pre-pilot balance in students' performance between treatment and control groups. The third- and fifth-grade cohorts had not reached the fifth grade before the pilot, so they do not have a baseline score.

Retention and *dropout* come from the annual School Census and are observed for every cohort in each year from 2015 to 2018. *High-school completion* also como from the School Census and *ENEM* participation and proficiency are observed in 2018 and exist only for the ninth-grade cohort, the only one to reach the end of high school within this window.

Source: Authors' elaboration based on the pilot roster merged with INEP's identified student-level administrative records (School Census, *Prova Brasil*, and *ENEM*). INEP performed the merge; the authors ran all analyses inside INEP's secure facility.

[25]Because a pilot cohort is only tested in *Prova Brasil* when it reaches a grade in which the exam is given, the timing of these outcomes differs across cohorts.

4 Identification Strategy

To estimate the causal impacts of the program on the outcomes of interest, we estimate the following reduced-form regression:

$$Y_{ism} = \alpha + \delta T_{sm} + \sum_{n=1}^{6} \gamma_n + \epsilon_{ism} \tag{1}$$

in which Y is the outcome variable (e.g., financial proficiency) of student i at school s in municipality m, T is a binary variable that is equal to 1 if the school s in municipality m was assigned to the program (treatment) and 0 otherwise'; γ_n are strata fixed effects, and ϵ is the idiosyncratic error.[26]

The parameter of interest, δ, measures the intention-to-treat (ITT), the treatment effect on students of schools randomly assigned to receive the treatment. The standard error is clustered at the school level, the randomization unit. This specification does not distinguish the effect between grades. To assess the program's effect only for elementary, only for middle school, and for each grade included in the pilot, we separately re-estimate Equation 1 for each one of these samples.[27] In addition to ITT effects, we follow Firpo et al. (2009) and estimate unconditional quantile intention-to-treat (UQITT) effects to assess whether the program has heterogeneous effects across the financial proficiency distribution.[28]

Because each table reports several outcomes at once, we adjust inference for multiple hypothesis testing with Romano–Wolf stepdown p-values (Romano and Wolf, 2005; Clarke et al., 2020), which control the probability of falsely rejecting at least one true null hypothesis when the outcomes of a table are tested jointly. While applications in this literature typically define families of related outcomes within a subsample (Frisancho, 2023a), we take a more conservative approach and treat all the regressions reported in each table, covering every outcome and every subsample, as a single family, computing the adjusted p-values with 1,000 bootstrap replications that resample school clusters within randomization strata. We report the significance level of every estimate according to this p-value, so the stars in the tables already reflect the multiple-hypothesis correction. The tables also report randomization-inference p-values (Young, 2019).

[26] Figure 1 shows that the cluster-randomized trial has six clusters.

[27] We also estimate standard errors clustering at the class level. The results are very similar and are reported in the online appendix.

[28] We estimate the UQITT in two steps. We first regress the outcome on a constant and the strata dummies and save the residuals; we then use the residuals as the dependent variable in a quantile regression on the treatment dummy. Since the estimates do not control for covariates, we call these unconditional quantile intention-to-treat effects. Estimates at the extreme quantiles are less precise, as the wider confidence intervals in Figure 2 show.

5 Results

5.1 Financial Proficiency, Savings and Consumption Index, and Behavioral Outcomes

Table 3 shows that the intervention has a positive and statistically significant impact of 0.07 of a standard deviation on students' financial proficiency, with the effect driven by the middle school grades. Larger impacts on students in more advanced grades are consistent with the hypothesis that financial education programs are more effective at increasing the financial proficiency of students who are more likely to make financial decisions in daily life (Bruhn et al., 2016; Frisancho, 2023a; Zia, 2023). School-based programs at large raise financial knowledge by 0.25 of a standard deviation on average (Kaiser and Menkhoff, 2020). Our estimate sits at the lower end of the wide range documented for elementary and middle school students, which runs from no detectable effect in Ghanaian primary and junior high schools (Berry et al., 2018) to 0.25 of a standard deviation among American fourth and fifth graders (Batty et al., 2020) and 0.46 among Belgian middle school students (De Beckker et al., 2021), although comparisons across programs are complicated by differences in intensity, content, delivery, and setting.

Our estimates also point to positive impacts on consumption and saving indices, the ones measuring students' hypothetical attitudes, particularly among fifth graders (Table 3). Students in treated classes report more prudent attitudes: they are less likely to endorse statements tolerating debt and impulsive spending, and more likely to endorse the importance of saving. Among fifth graders, the program raised the consumption index by 0.12 of a standard deviation and the savings index by 0.09 (column 5). Among middle school students (column 3), the effect on the consumption index remains statistically significant, although smaller at 0.08 of a standard deviation, while the effect on the savings index, at 0.04, is not statistically significant, suggesting that the program did less to shift the reported savings attitudes of older students.

The larger point estimates among fifth graders might be associated with a few factors. Younger students might be less overconfident than teenagers, making them more receptive to the program's message. In addition, it might be easier to shift the reported attitudes of younger students, whose financial habits are not consolidated yet.

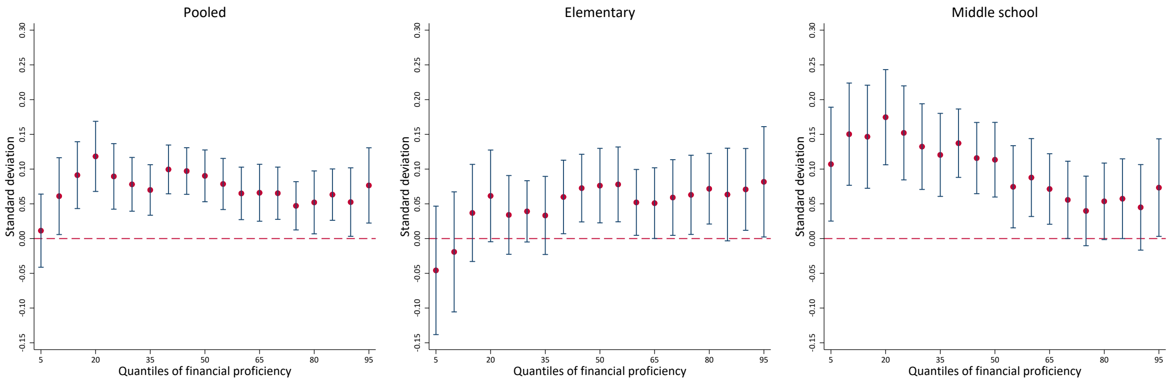
Table 3: ITT estimates for financial proficiency and savings and consumption index

	(1) Pooled	(2) Elementary	(3) Middle school	(4) Third grade	(5) Fifth grade	(6) Seventh grade	(7) Ninth grade
Financial proficiency							
Treatment	0.073**	0.047	0.099**	0.053	0.043	0.090*	0.106*
	(0.034)	(0.046)	(0.042)	(0.077)	(0.046)	(0.048)	(0.055)
pvalue RW	0.038	0.428	0.025	0.597	0.482	0.072	0.072
pvalue RI	0.036	0.302	0.017	0.521	0.341	0.063	0.064
N. obs	14655	7641	7014	3635	4006	3640	3374
R-squared	0.167	0.150	0.188	0.127	0.173	0.208	0.168
Consumption index							
Treatment	0.094***	0.122***	0.082**		0.122***	0.080*	0.084**
	(0.026)	(0.042)	(0.031)		(0.042)	(0.041)	(0.037)
pvalue RW	0.001	0.005	0.013		0.005	0.068	0.027
pvalue RI	0.000	0.005	0.011		0.009	0.060	0.033
N. obs	10220	3648	6572		3648	3409	3163
R-squared	0.089	0.119	0.076		0.119	0.100	0.055
Saving index							
Treatment	0.055**	0.087**	0.044		0.087**	0.066	0.018
	(0.025)	(0.041)	(0.029)		(0.041)	(0.040)	(0.038)
pvalue RW	0.036	0.042	0.177		0.042	0.122	0.597
pvalue RI	0.035	0.046	0.143		0.036	0.106	0.663
N. obs	10022	3549	6473		3549	3368	3105
R-squared	0.046	0.096	0.025		0.096	0.032	0.019

Notes: Clustered standard errors at the school level in parentheses. RW denotes Romano-Wolf stepdown p-values (1,000 bootstrap replications resampling school clusters within randomization strata), adjusting for the joint testing of the 19 regressions reported in this table (every outcome in every subsample) as a single family, and RI denotes randomization inference (1,000 repetitions); both are reported below the standard error. Stars denote significance levels based on the Romano-Wolf adjusted p-value: * 10%, ** 5%, *** 1%. Financial proficiency is computed by CAEd from students' answers to a standardized test scored with Item Response Theory. The consumption index is the first principal component of students' answers to seven agreement-scale statements, and the savings index the first principal component of eight such statements; the statements and their weights are reported in Table A.3. All three indices are normalized to standard-deviation units, so treatment effects are read in standard deviations. The questionnaire on hypothetical consumption and saving attitudes was not administered in the third grade. For these two outcomes, column (4) is therefore empty and column (2) reports the same estimates as column (5), the fifth grade.
Source: Authors' estimates based on the pilot questionnaires applied by CAEd to students at the end of the 2015 school year.

To investigate whether the intervention had distributional impacts on financial proficiency, we estimate unconditional quantile intention-to-treat (UQITT) effects, as described in section 4. Figure 2 displays the UQITT estimates at nineteen quantiles of the financial proficiency distribution, for the pooled sample and separately for elementary and middle school students. Among elementary school students, the estimates become positive and statistically significant, at about 0.05 of a standard deviation, from the 45th percentile onward: the program benefited a subset of elementary school students, while having no detectable effect on the bottom 40 percent. Among middle school students, the UQITT and ITT estimates are similar, suggesting a more homogeneous effect across the distribution in higher grades.

Figure 2: Unconditional Quantile ITT estimates for financial proficiency - 95% CI



Note: Financial proficiency is computed by CAEd from students' answers to a standardized test scored with Item Response Theory. The index is normalized to standard-deviation units, so treatment effects are read in standard deviations. We first run this financial proficiency score on the treatment dummy, the six strata dummies, and a constant. We then take the residuals from this estimation and run a simultaneous quantile regression for each quantile shown in the Figure, with the dependent variable being the residual from the first regression and the treatment dummy as the independent variable. 1000 bootstrap replications. The three panels report the pooled sample, elementary school students (third and fifth grades), and middle school students (seventh and ninth grades).
Source: Authors' estimates based on the pilot financial-literacy test applied by CAEd to students at the end of the 2015 school year.

We further explore whether the program affected students' behavioral outcomes by examining more objective measures of behavioral change. Table 4 shows that fifth graders are 5.7 pp (11.6 percent over the control mean) more likely to talk with their parents about money or family expenses, and 5.6 pp (28.4 percent) more likely to talk about money with their friends. There is also some evidence that middle school students became more likely to talk about money with their friends, by 2.3 pp (8.4 percent), although the effect is only significant at the 10 percent level. These conversations are a channel worth taking seriously: in Peru, school-based financial education spilled over to the credit behavior of the students' parents, especially in poorer households (Frisancho, 2023b). We find no evidence that the pilot affected the other behavioral outcomes: piggy bank use, the probability of receiving an allowance, and the use of financial services such as credit or debit cards. The last result is unsurprising in our context, given that most students are young and do not deal with financial transactions in their daily lives.

Our estimates show that the program had larger impacts on financial proficiency in later grades and larger effects on attitudes and behavior in earlier grades. The results then suggest that the association between financial proficiency and behavioral change might not be straightforward. Individuals might have formal knowledge across different subjects, such as financial proficiency, but the application of that knowledge in real life can be hindered by the institutional environment they are embedded in (Carpena and Zia, 2020). To formally test the hypothesis that learning mediates change in behavior, we follow Imai et al. (2011) and decompose the treatment effect estimates into the average causal mediation effect (ACME) and the average causal direct effect

Table 4: ITT estimates for behavioral outcomes

	(1) Pooled	(2) Elementary	(3) Middle school
Talks about money or expenses with parents			
Treatment	0.012	0.057***	-0.012
	(0.010)	(0.019)	(0.012)
pvalue RW	0.489	0.002	0.530
pvalue RI	0.238	0.005	0.295
N. obs	10748	3875	6873
R-squared	0.006	0.003	0.001
Talks about money with friends			
Treatment	0.035***	0.056***	0.023*
	(0.009)	(0.014)	(0.011)
pvalue RW	0.001	0.001	0.081
pvalue RI	0.000	0.000	0.059
N. obs	10725	3864	6861
R-squared	0.008	0.012	0.005
Piggy bank use			
Treatment	0.019	0.021	0.017
	(0.011)	(0.017)	(0.013)
pvalue RW	0.175	0.489	0.489
pvalue RI	0.088	0.225	0.209
N. obs	10688	3866	6822
R-squared	0.026	0.037	0.010
Use of financial services			
Treatment	0.004	-0.003	0.006
	(0.010)	(0.016)	(0.013)
pvalue RW	0.841	0.841	0.816
pvalue RI	0.754	0.858	0.658
N. obs	10682	3862	6820
R-squared	0.003	0.004	0.006
Receiving an allowance			
Treatment	0.013	0.017	0.010
	(0.008)	(0.014)	(0.010)
pvalue RW	0.271	0.489	0.530
pvalue RI	0.136	0.216	0.309
N. obs	10699	3862	6837
R-squared	0.009	0.012	0.007

Notes: Clustered standard errors at the school level in parentheses. RW denotes Romano-Wolf stepdown p-values (1,000 bootstrap replications resampling school clusters within randomization strata), adjusting for the joint testing of the 15 regressions reported in this table (every outcome in every subsample) as a single family, and RI denotes randomization inference (1,000 repetitions); both are reported below the standard error. Stars denote significance levels based on the Romano-Wolf adjusted p-value: * 10%, ** 5%, *** 1%. Talking about money or expenses with parents equals 1 if the student reports discussing household expenses or talking about money with their parents, and 0 otherwise; the remaining variables equal 1 if the student talks about money with friends, keeps a piggy bank, has access to at least one financial service (bank account, debit card, or credit card), or receives an allowance from their parents, and 0 otherwise. The sample is restricted to fifth-, seventh-, and ninth-grade students.
Source: Authors' estimates based on the pilot questionnaires applied by CAEd to students at the end of the 2015 school year.

(ADE).[29] The goal of this exercise is to assess the share of the change in behavioral outcomes that could be attributed to improvements in financial proficiency (ACME) and the share that comes directly from the program (ADE).[30]

[29] See the online appendix for a formal discussion of Causal Mediation Effects.

[30] The decomposition of ITT effects on ACME and ADE relies on the sequential ignorability assumption. The first part of the assumption assumes that, conditioned on pre-treatment confounders, the treatment assignment is orthogonal to the potential outcomes and the mediator outcome (financial proficiency). The second part of the assumption implies that the observed mediator is ignorable given the actual treatment status and pre-treatment

According to the estimates in Table A.9, financial proficiency mediates part of the improvement in students' hypothetical attitudes toward consumption: among middle school students, the mediated component accounts for about half of the total effect. For elementary school students, by contrast, every mediated component is statistically indistinguishable from zero, so the effects on attitudes and on talking about money with parents and friends are direct. The same holds for the behavioral outcomes at both school levels, where the mediated components are negligible. In other words, the ADE and ACME estimates suggest that financial education programs affect students' attitudes and behavior through pathways other than the accumulation of formal knowledge. Overall, the estimates do not support the hypothesis that changes in financial proficiency are a necessary condition for changes in individuals' behavioral outcomes, a finding consistent with Carpena and Zia (2020) and De Beckker et al. (2021).

Finally, we test whether the gains in financial proficiency and the more prudent attitudes toward saving and consumption, which might indicate that students became more forward-looking, translate into higher investment in human capital. If that were the case, we should see lower repetition and dropout and higher proficiency in Portuguese and math.

Table 5 shows no systematic evidence that the program affected these outcomes. These outcomes are, however, observed over a relatively short window, one to three years after a low-intensity intervention, so changes that take longer to materialize, especially at these young ages, may fall outside what our data can capture.

5.2 Robustness Check

As discussed in section 3, the student-level merge with INEP's records recovers pre-treatment proficiency scores in reading and math for the seventh- and ninth-grade cohorts, and Table A.8 shows that ninth-grade control students outperformed treated ones in reading before the pilot. Since principals, and not chance, chose the participating classes, this imbalance could bias our estimates. We therefore re-estimate Equation 1 for these two cohorts controlling for their fifth-grade proficiency scores.[31]

Controlling for past performance leads to a substantial gain in precision. Among ninth graders, the point estimate for financial proficiency rises from 0.107 to 0.149 of a standard deviation; and among seventh graders, it rises slightly from 0.09 to 0.098 (Table A.10). The pattern suggests that ignoring the pre-treatment imbalance would underestimate the program's impact on financial

confounders (Imai et al., 2011). This second part of the assumption is the strongest, even in the presence of random assignment. It posits that, conditional on the actual treatment (not the treatment assignment) and a vector of pre-treatment confounders, the outcome is independent of the mediator. Note that in our case, this implies the assumption that attitudinal and behavioral outcomes are orthogonal to financial proficiency, given the actual treatment and after controlling for a vector of school characteristics. Since we did not design the experiment to obtain clean estimates of the ACME and ADE, we interpret our results as suggestive evidence.

[31] These are the only two cohorts with a pre-treatment performance score: seventh graders sat the fifth-grade test in 2013 and ninth graders in 2011.

Table 5: ITT estimates for human capital accumulation outcomes

	Third grade (1)	Fifth grade (2)	Seventh grade (3)	Ninth grade (4)
Panel A. Grade progression (School Census, cumulative 2015–2018)				
Repetition	0.017	0.029**	-0.008	0.018
	(0.017)	(0.014)	(0.016)	(0.018)
Dropout	0.005	0.011	0.002	0.020
	(0.006)	(0.010)	(0.012)	(0.019)
Panel B. Test scores (Prova Brasil)				
Assessed in	*2017*	*2015*	*2017*	*2015*
Portuguese	-0.029	-0.017	-0.005	0.028
	(0.047)	(0.046)	(0.034)	(0.036)
Math	-0.066	-0.033	-0.009	-0.020
	(0.053)	(0.049)	(0.038)	(0.039)
Panel C. End of high school (ninth-grade cohort assessed in 2018)				
Graduated from high school				-0.019
				(0.023)
Participated in ENEM				-0.012
				(0.023)
Portuguese ENEM				-0.025
				(0.031)
Math ENEM				0.002
				(0.045)
Obs.	4650	4755	3137	2851

Notes: Clustered standard errors at the school level in parentheses. Stars denote significance levels based on conventional p-values: * 10%, ** 5%, *** 1%. The estimates were produced inside INEP's secure facility, where the bootstrap-based Romano-Wolf were not run; since the estimates are null, adjusting for multiple hypothesis testing would only reinforce the conclusions. Each dependent variable is regressed on treatment assignment, with one specification per cohort. Columns (1) and (2) include strata fixed effects only, since the third- and fifth-grade cohorts have no pre-pilot test score. Columns (3) and (4) add the student's pre-pilot fifth-grade *Prova Brasil* performance, taken in 2013 by the seventh-grade cohort and in 2011 by the ninth-grade cohort, which controls for the pre-treatment imbalance documented in Table A.8.
Timing: repetition and dropout (Panel A) are cumulative over 2015–2018. In Panel B, the "Assessed in" row gives the year each cohort sits *Prova Brasil*: the fifth- and ninth-grade cohorts are assessed in the pilot year (2015, contemporaneous), while the third- and seventh-grade cohorts are assessed two years later (2017), when they reach the next assessment grade (see Table 2). Panel C outcomes refer to 2018 and exist only for the ninth-grade cohort.
Source: Authors' estimates based on the pilot roster merged with INEP's identified student-level administrative records (School Census, *Prova Brasil*, and ENEM). INEP performed the merge; the authors ran all estimates inside INEP's secure facility.

proficiency, particularly for ninth graders. Controlling for past performance does not change the estimates for consumption and saving attitudes.

5.3 Implementation and IV Results

As discussed in subsection 2.3, we found evidence of contamination, especially in the control group. Because exposure to financial education content varied across classes within schools in both arms, the ITT compares assignment rather than exposure. To recover the effect of actual exposure, we use the random assignment as an instrument for exposure, following Frisancho (2023a).

We measure exposure using two definitions, regardless of the arm to which the school was assigned. The conservative one is defined at the class level: it sets $Z = 1$ when a teacher reports delivering financial education classes.[32] For the classes whose teachers did not answer the teacher questionnaire, we set $Z = 1$ if more than half of the students from those classes report having had such classes. The broad one is defined at the student level: it sets $Z = 1$ when a teacher of the student's class reports delivering the financial education classes, as in the conservative definition, or when the student themselves reports having had them. Under the conservative definition, 97.8% of students in treated classes and 11.1% in control classes are classified as exposed; under the broad definition, 97.3% and 21.5%, respectively.

We estimate the local average treatment effect (LATE) by two-stage least squares. The first stage is

$$Z_{ism} = \pi_0 + \pi_1 T_{ism} + \sum_{n=1}^{6} \gamma_n + \nu_{ism} \tag{2}$$

and the second stage is

$$Y_{ism} = \lambda_0 + \lambda_1 \widehat{Z}_{ism} + \sum_{n=1}^{6} \gamma_n + \omega_{ism} \tag{3}$$

where Z is the exposure of student i's class under the conservative definition and the student's own exposure under the broad one, T is the random assignment of Equation 1, and the remaining notation follows that equation. Standard errors are clustered at the school level, the randomization unit. The parameter of interest, λ_1, is the LATE: the average effect of exposure among compliers, that is, the classes (under the conservative definition) or students (under the broad definition) that were exposed to the program because their school was assigned to the treatment group, and would not have been exposed otherwise.

Table 6 reports the LATE estimates next to the ITT under both definitions of exposure.[33] For financial proficiency the differences are moderate: among middle school students, the effect rises from 0.099 of a standard deviation to 0.112 under the conservative definition and 0.135 under the broad one. For the attitude indices in grade 5, the correction is substantial: under the broad definition, the effect on the consumption index rises from 0.122 to 0.202 of a standard deviation, and on the saving index from 0.087 to 0.143. Exposure to the program therefore had modest effects on financial proficiency even among the classes that delivered the content, while its effects on the reported attitudes of younger students are sizeable once imperfect compliance is taken into account; since the broad definition relies on students' own reports, these larger estimates are best read as upper bounds.

[32] If the teacher answered the question "*How many financial education classes did you teach?*", with answer categories running from one to eight or more, or the question "*What share of the student textbook did you cover?*", with categories running from below 40% to above 80%.

[33] The first-stage estimates behind each LATE are reported in Table A.11.

Table 6: Intention-to-treat (ITT) and local average treatment effect (LATE) estimates

	Pooled			Elementary			Middle school		
	ITT	LATE (i)	LATE (ii)	ITT	LATE (i)	LATE (ii)	ITT	LATE (i)	LATE (ii)
Financial proficiency									
Treatment	0.073**	0.083**	0.098**	0.047	0.055	0.063	0.099**	0.112***	0.135**
	(0.034)	(0.039)	(0.046)	(0.046)	(0.053)	(0.061)	(0.042)	(0.048)	(0.058)
pvalue RW	0.023	0.023	0.023	0.165	0.165	0.165	0.011	0.009	0.012
N. obs	14655	14655	14655	7641	7641	7641	7014	7014	7014
R-squared	0.167	0.166	0.162	0.150	0.150	0.148	0.188	0.189	0.179
Consumption index									
Treatment	0.094***	0.109***	0.136***	0.122***	0.150***	0.202***	0.082***	0.093***	0.112***
	(0.026)	(0.031)	(0.039)	(0.042)	(0.052)	(0.070)	(0.031)	(0.035)	(0.043)
pvalue RW	0.001	0.001	0.001	0.003	0.003	0.003	0.006	0.006	0.006
N. obs	10220	10220	10220	3648	3648	3648	6572	6572	6572
R-squared	0.089	0.087	0.081	0.119	0.117	0.110	0.076	0.076	0.070
Saving index									
Treatment	0.055**	0.063**	0.079**	0.087**	0.107**	0.143**	0.044	0.050	0.060
	(0.025)	(0.029)	(0.037)	(0.041)	(0.051)	(0.069)	(0.029)	(0.033)	(0.040)
pvalue RW	0.023	0.023	0.023	0.023	0.023	0.023	0.103	0.103	0.103
N. obs	10022	10022	10022	3549	3549	3549	6473	6473	6473
R-squared	0.046	0.046	0.043	0.096	0.095	0.092	0.025	0.025	0.023

Notes: Clustered standard errors at the school level in parentheses. RW denotes Romano-Wolf stepdown p-values (1,000 bootstrap replications resampling school clusters within randomization strata), adjusting for the joint testing of the 27 regressions reported in this table (every outcome, subsample, and estimator) as a single family, reported below the standard error. Stars denote significance levels based on the Romano-Wolf adjusted p-value: * 10%, ** 5%, *** 1%. ITT is the reduced-form effect of random assignment; LATE instruments actual program exposure (Z) with random assignment. Exposure is measured under two definitions: (i) the conservative one, defined at the class level, which requires a teacher of the class to report having taught the financial education classes or covered part of the textbook, and, for the classes whose teachers did not answer their pilot questionnaire, that more than half of the students report having had such classes; and (ii) the broad one, defined at the student level, which sets $Z = 1$ when a teacher of the class reports delivering the classes or the student themselves reports having had them. Both are defined in the text. All three indices are normalized to standard-deviation units. The consumption and savings indices are unavailable for third graders, who did not answer these items. Therefore, elementary estimates represent the fifth grade, while middle school estimates represent the seventh and ninth grades.
Source: Authors' estimates based on the pilot questionnaires applied by CAEd to students at the end of the 2015 school year.

6 Conclusion

In this paper, we use a cluster randomized controlled trial to evaluate the impacts of a financial education pilot program in Brazilian elementary schools during the academic year of 2015. The pilot was implemented in 101 municipal schools and involved approximately 9,000 students. Our main findings showed that the program improved students' financial proficiency, their reported attitudes toward consumption and saving, and some behavioral outcomes, such as talking about money with parents and friends. We find strong indications of heterogeneous effects though. While proficiency gains were stronger among middle school students, changes in attitudes and behavioral outcomes stood out among elementary education students.

We tested the overlooked hypothesis in this literature that poses that change in formal knowledge precedes changes in actual behavior. We use a causal mediation effect analysis to test this hypothesis. Overall, our results seem to reject the hypothesis that claims that changes in knowledge are a necessary condition to change individuals' habits.

Finally, we used student test score data in math and reading in different school grades to investigate the program's impacts on contemporary learning outcomes and on human capital accumulation. We do not find evidence that the program impacted grade progression, retention, dropout rates, or learning outcomes. Our reading of the results is that the program did not affect students' intertemporal decision-making, confirming the preliminary results used by the implementing partner who decided to scale the program down in its current format.

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A Tables

Table A.1: Program implementation, 2015

	Pooled			Elementary			Middle		
	C	T	Diff	C	T	Diff	C	T	Diff
Panel A. Implementation according to students (% of students)									
Did you have financial education classes?									
Yes	16.3	66.8	-50.5***	21.3	73.9	-52.6***	13.7	63.1	-49.4***
No	59.0	10.1	48.9***	57.2	8.1	49.1***	60.0	11.2	48.8***
Did not answer	24.7	23.1	1.6	21.5	18.0	3.5*	26.3	25.7	0.6
When did you receive the financial education textbook?									
Beginning of the year	10.0	16.1	-6.1***	10.9	20.5	-9.6***	9.5	13.7	-4.2***
By the middle of the year	5.6	41.9	-36.3***	7.4	46.9	-39.6***	4.7	39.3	-34.7***
In the end of the year	4.8	10.9	-6.1***	6.8	9.1	-2.3*	3.8	11.9	-8.1***
Haven't received	11.1	5.2	6.0***	11.5	4.6	6.9***	11.0	5.5	5.5***
Did not answer	68.5	25.9	42.6***	63.4	18.8	44.6***	71.1	29.6	41.5***
Has your teacher used the financial education textbook?									
Received and used	7.7	57.9	-50.2***	10.9	64.2	-53.4***	6.1	54.5	-48.5***
Received but haven't used	5.7	9.1	-3.4***	6.4	10.2	-3.8***	5.3	8.6	-3.3***
Haven't received	17.5	7.2	10.4***	18.4	6.8	11.6***	17.1	7.4	9.7***
Did not answer	69.1	25.8	43.3***	64.3	18.7	45.6***	71.6	29.5	42.0***
Observations	6,973	6,934		2,375	2,380		4,598	4,554	
Panel B. Implementation according to teachers (% of teachers)									
Received training to use the textbook									
Yes	2.3	72.3	-69.9***	3.7	73.6	-69.9***	0.7	71.2	-70.4***
No	15.6	23.6	-7.9*	16.0	26.4	-10.5**	15.2	21.2	-5.9
Did not answer	82.1	4.2	77.9***	80.4	0.0	80.4***	84.1	7.7	76.4***
Received the teacher book									
Yes	3.7	95.0	-91.4***	5.5	97.1	-91.6***	1.4	93.3	-91.8***
No	15.3	2.4	12.9***	14.1	2.9	11.2***	16.7	1.9	14.7***
Did not answer	81.1	2.6	78.4***	80.4	0.0	80.4***	81.9	4.8	77.1***
Student books were distributed to the students									
Yes	2.3	93.7	-91.4***	3.7	97.7	-94.0***	0.7	90.4	-89.7***
No	16.3	3.4	12.9***	16.0	1.7	14.2***	16.7	4.8	11.9***
Did not answer	81.4	2.9	78.5***	80.4	0.6	79.8***	82.6	4.8	77.8***
Used the student book									
Yes	3.7	90.3	-86.7***	4.3	96.0	-91.7***	2.9	85.6	-82.7***
No	14.3	6.8	7.5***	14.7	3.4	11.3***	13.8	9.6	4.2
Did not answer	82.1	2.9	79.2***	81.0	0.6	80.4***	83.3	4.8	78.5***
Number of financial education classes taught									
1	5.6	0.3	5.4***	5.5	0.6	4.9**	5.8	0.0	5.8**
2	1.3	3.1	-1.8	1.8	3.4	-1.6	0.7	2.9	-2.2
3	0.3	2.4	-2.0**	0.6	2.9	-2.3	0.0	1.9	-1.9**
4	1.0	7.1	-6.1***	1.2	3.4	-2.2	0.7	10.1	-9.4***
5	0.3	8.6	-8.3***	0.6	6.3	-5.7***	0.0	10.6	-10.6***
6	0.0	10.2	-10.2***	0.0	6.9	-6.9***	0.0	13.0	-13.0***
7	1.0	11.5	-10.5***	1.2	12.6	-11.4***	0.7	10.6	-9.9***
8 or more	1.7	52.4	-50.7***	1.8	62.1	-60.2***	1.4	44.2	-42.8***
Did not answer	88.7	4.5	84.3***	87.1	1.7	85.4***	90.6	6.7	83.8***
When in the year was the subject taught?									
First semester	5.6	23.3	-17.7***	6.1	23.6	-17.4***	5.1	23.1	-18.0***
Second semester	2.0	41.4	-39.4***	2.5	32.2	-29.7***	1.4	49.0	-47.6***
All year	2.3	29.8	-27.5***	3.1	41.4	-38.3***	1.4	20.2	-18.7***
Did not answer	90.0	5.5	84.5***	88.3	2.9	85.5***	92.0	7.7	84.3***
Share of the textbook covered									
Less than 40%	3.7	6.8	-3.2	3.1	2.9	0.2	4.3	10.1	-5.7*
40%–60%	1.3	28.0	-26.7***	2.5	26.4	-24.0***	0.0	29.3	-29.3***
60%–80%	4.0	36.1	-32.1***	3.7	38.5	-34.8***	4.3	34.1	-29.8***
More than 80%	1.0	25.1	-24.1***	1.8	31.6	-29.8***	0.0	19.7	-19.7***
Did not answer	90.0	3.9	86.1***	89.0	0.6	88.4***	91.3	6.7	84.6***
Observations	301	382		163	174		138	208	

Notes: C and T are control and treatment means and Diff is the difference between them. Significance is from a regression of the indicator on treatment status and the strata dummies, with standard errors clustered at the school level; stars denote significance levels: * 10%, ** 5%, *** 1%. In Panel A, the unit of observation is the student in grades 5, 7, or 9, for which the data are available: Elementary is grade 5 and Middle is grades 7 and 9. In Panel B, the unit of observation is the teacher in charge of a pilot class: Elementary covers grades 3 and 5, and Middle grades 7 and 9.

Source: Authors' estimates based on the pilot questionnaires applied by CAEd to students and teachers at the end of the 2015 school year.

Table A.2: Descriptors and skills on the financial knowledge test

		Third grade	Fifth grade	Seventh grade	Ninth grade
D01	Being able to identify the subject of texts whose topic explores socially responsible attitudes towards the environment	yes	yes	yes	yes
D02	Being able to find information in texts about consumption – light, water, telephone bills, among others	yes	yes	no	no
D03	Being able to identify the purpose of texts and text formats that include expenses, consumption, spending	yes	yes	no	no
D04	Being able to recognize the purpose of text genres related to finances – receipts, checks, invoices	no	yes	yes	yes
D05	Being able to recognize situations in which concepts related to finances are present: savings, expenses, consumption, spending, waste, risk, return, financial planning, and investment, among others.	no	no	yes	yes
D06	Being able to identify situations related to financially responsible attitudes	no	yes	yes	yes
D07	Being able to find information in graphs and tables that contain data related to finances (purchases, sales, spending)	yes	no	no	no
D08	Being able to find information in texts that circulate in the financial world: classified ads, news features, among others	yes	yes	yes	yes
D09	Being able to estimate values and/or procedures necessary for financial projects	yes	yes	yes	yes
D10	Being able to distinguish remunerated from non-remunerated work.	yes	yes	no	no
D11	Being able to identify the origin and destination of varied products and/or those that can be recycled.	yes	no	no	no
D12	Being able to recognize socially responsible situations related to public and private spaces.	no	no	no	yes
D13	Being able to identify advantages, disadvantages, and risks of cash and credit sales.	no	yes	no	yes
D14	Being able to find implicit information in media texts that are relevant for decision-making in finances.	no	yes	yes	yes

Notes: Each row is a skill descriptor assessed by the financial literacy test; "yes" indicates the descriptor is assessed in that grade. The test content is adapted to each grade, so not every descriptor appears in every grade.
Source: Authors' elaboration based on the financial literacy test applied by CAEd at the end of the 2015 school year.

Table A.3: Items used to construct the consumption and savings attitude indices

Statement	Answers' scale				Weight
	Totally agree	Agree	Disagree	Totally disagree	
Consumption questions					
I buy what I want, then I see how I can pay	1	2	3	4	0.45
I see no problem in owing money	1	2	3	4	0.44
If the brand is famous the product is of high quality	1	2	3	4	0.32
The best product is always the most expensive	1	2	3	4	0.33
I plan before spending my money	4	3	2	1	0.10
It is worthless to plan because the money comes from luck	1	2	3	4	0.43
Buy what I want is more important than planning	1	2	3	4	0.45
Saving questions					
I think that saving money is important to avoid problems in the future	4	3	2	1	0.41
I feel safer when I can save some money	4	3	2	1	0.39
Saving some money is important to avoid debt	4	3	2	1	0.42
Buying everything I want is more important than putting the money together	1	2	3	4	0.24
Avoiding waste is also a way to save money	4	3	2	1	0.38
I try to use the products for longer	4	3	2	1	0.35
Whenever I can, I save money	4	3	2	1	0.37
I would rather spend the change on something I want than save the money for later	1	2	3	4	0.19

Notes: The table lists every item used to build the two indices: seven for consumption and eight for savings. Each item was printed in the questionnaire as a statement, and students chose the option they most agreed with on a four-point scale. The figures under *Answers' scale* show how each option was coded: because the statements are worded in both directions, the coding is reversed for the positively worded items so that 4 always corresponds to the most prudent response.
Each index is the first principal component of the items in its block, standardized to mean zero and unit variance. *Weight* reports the weight the principal component places on each item.
Source: Authors' elaboration based on the pilot student questionnaire applied by CAEd at the end of the 2015 school year.

Table A.4: Balance test for students

		Third grade			Fifth grade			Seventh grade			Ninth grade		
		C	T	Diff	C	T	Diff	C	T	Diff	C	T	Diff
Participation: financial literacy assessment	Mean	77.4	79.0	-1.7	83.9	84.6	-0.7	76.7	78.3	-1.6	77.0	74.4	2.6
	SE	(1.8)	(1.9)		(1.4)	(1.4)		(2.1)	(1.9)		(1.8)	(2.3)	
	Obs	2380	2270		2375	2380		2376	2321		2222	2233	
Response: socioeconomic questionnaire	Mean	65.3	65.6	-0.3	97.4	99.1	-1.7	99.6	99.7	-0.2	97.7	99.3	-1.6
	SE	(2.2)	(2.9)		(1.5)	(0.3)		(0.1)	(0.1)		(1.0)	(0.2)	
	Obs	1808	1756		1992	2014		1822	1818		1712	1638	
Access to paved street	Mean	68.7	66.8	1.9	67.6	67.4	0.2	71.5	69.4	2.1	73.7	75.1	-1.4
	SE	(3.7)	(3.2)		(2.6)	(2.4)		(2.9)	(2.9)		(2.9)	(3.0)	
	Obs	1087	1028		1914	1983		1804	1802		1662	1616	
Access to electricity	Mean	99.6	99.1	0.5	95.9	96.0	-0.1	97.6	97.6	-0.0	97.9	97.0	0.9
	SE	(0.2)	(0.4)		(0.7)	(0.5)		(0.5)	(0.4)		(0.3)	(0.5)	
	Obs	1041	1005		1914	1980		1800	1803		1668	1621	
Access to piped water	Mean	96.7	96.3	0.4	96.1	96.2	-0.2	96.5	96.7	-0.3	97.3	96.8	0.5
	SE	(1.0)	(0.9)		(0.7)	(0.6)		(0.7)	(0.4)		(0.7)	(0.5)	
	Obs	1060	1023		1925	1988		1806	1808		1667	1618	
Access to garbage collection	Mean	91.5	93.3	-1.8	85.7	84.3	1.4	85.2	88.7	-3.5	90.2	91.2	-1.1
	SE	(2.0)	(1.5)		(1.4)	(1.3)		(2.0)	(1.2)		(1.4)	(1.0)	
	Obs	1050	1002		1907	1961		1794	1796		1668	1609	
Beneficiary of Bolsa Família	Mean	36.2	36.1	0.2	45.4	44.9	0.4	40.9	42.9	-1.9	38.3	35.8	2.6*
	SE	(3.1)	(2.9)		(2.8)	(2.7)		(2.9)	(2.9)		(3.1)	(2.6)	
	Obs	1082	1059		1905	1958		1790	1788		1648	1604	
Adequate age for their grade	Mean	87.9	86.6	1.3	82.3	80.6	1.7	79.9	80.1	-0.2	81.2	81.1	0.0
	SE	(1.4)	(1.7)		(1.6)	(1.5)		(2.1)	(1.8)		(1.7)	(2.1)	
	Obs	1038	997		1902	1952		1781	1781		1636	1602	
Mother education/Legal guardian: incomplete elementary school	Mean	26.1	28.2	-2.1	26.3	25.2	1.1	25.1	26.9	-1.8	24.1	22.8	1.3
	SE	(2.0)	(2.1)		(1.5)	(1.7)		(1.5)	(1.6)		(1.6)	(1.4)	
	Obs	1081	1050		1272	1333		1347	1317		1364	1335	
Mother education/Legal guardian: incomplete high school	Mean	20.9	20.9	0.0	18.1	18.8	-0.7	22.3	18.4	3.9*	18.6	20.1	-1.5
	SE	(1.7)	(1.2)		(1.2)	(1.2)		(1.4)	(1.4)		(1.0)	(1.2)	
	Obs	1081	1050		1272	1333		1347	1317		1364	1335	
Mother education/Legal guardian: at least high school degree	Mean	53.0	51.0	2.1	55.7	56.0	-0.4	52.6	54.7	-2.1	57.3	57.0	0.3
	SE	(2.6)	(2.3)		(1.9)	(2.0)		(2.0)	(2.1)		(1.7)	(1.8)	
	Obs	1081	1050		1272	1333		1347	1317		1364	1335	
Gender: male	Mean				51.3	50.1	1.2	48.0	48.1	-0.1	49.5	49.6	-0.1
	SE				(1.3)	(1.3)		(1.5)	(1.0)		(1.1)	(1.5)	
	Obs				1905	1964		1798	1799		1660	1614	
Color: white	Mean				40.6	37.3	3.2	36.5	34.9	1.6	37.0	35.5	1.5
	SE				(2.5)	(2.5)		(2.9)	(2.4)		(2.9)	(2.7)	
	Obs				1888	1945		1771	1787		1634	1585	

Notes: C = control, T = treatment; Diff is the difference between the two group means. Means and differences are in percentage points, with the standard error of each mean in parentheses. Group means and their standard errors come from a regression on a constant within each group; each difference is tested with a regression of the variable on treatment status and the strata dummies, with standard errors clustered at the school level; stars denote significance levels from that test: * 10%, ** 5%, *** 1%. The first two rows report participation in the financial literacy assessment and, among participants, the response rate to the socioeconomic questionnaire, answered by the legal guardians in the third grade.
Source: Authors' estimates based on the pilot socioeconomic questionnaire applied by CAEd at the end of the 2015 school year.

Table A.5: Balance test for teachers

		Third grade			Fifth grade			Seventh grade			Ninth grade		
		C	T	Diff	C	T	Diff	C	T	Diff	C	T	Diff
Gender: male	Mean	7.2	9.9	-2.6	10.0	20.9	-10.9**	48.5	33.6	14.8*	38.6	44.6	-6.0
	SE	(2.9)	(3.3)		(3.4)	(4.2)		(5.8)	(4.3)		(5.8)	(4.7)	
	Obs	83	81		80	91		66	107		70	101	
Age: less than 35 years	Mean	27.7	26.8	0.9	26.3	30.4	-4.2	35.8	33.6	2.2	26.8	31.7	-4.9
	SE	(4.9)	(4.9)		(5.0)	(5.3)		(6.2)	(4.6)		(5.3)	(4.4)	
	Obs	83	82		80	92		67	107		71	101	
Age: 36 to 50 years	Mean	50.6	59.8	-9.2	60.0	53.3	6.7	44.8	54.2	-9.4	53.5	50.5	3.0
	SE	(5.5)	(5.4)		(5.5)	(5.5)		(6.1)	(5.1)		(5.9)	(4.8)	
	Obs	83	82		80	92		67	107		71	101	
Age: older than 51 years	Mean	21.7	13.4	8.3	13.8	16.3	-2.6	17.9	11.2	6.7	19.7	16.8	2.9
	SE	(4.6)	(3.8)		(3.9)	(3.9)		(4.6)	(3.0)		(4.7)	(3.8)	
	Obs	83	82		80	92		67	107		71	101	
Color: white	Mean	45.8	42.7	3.1	48.8	38.0	10.7*	43.3	35.5	7.8*	46.5	38.6	7.9
	SE	(5.5)	(5.5)		(5.6)	(5.7)		(5.9)	(5.4)		(5.9)	(5.6)	
	Obs	83	82		80	92		67	107		71	101	
Experience: up to 5 years	Mean	19.3	17.1	2.2	11.3	19.6	-8.3	17.9	15.0	3.0	14.1	14.9	-0.8
	SE	(4.4)	(4.2)		(3.6)	(4.0)		(4.7)	(3.1)		(4.2)	(3.6)	
	Obs	83	82		80	92		67	107		71	101	
Experience: 6 to 15 years	Mean	41.0	46.3	-5.4	42.5	41.3	1.2	40.3	57.0	-16.7**	52.1	45.5	6.6
	SE	(5.4)	(5.5)		(5.6)	(5.1)		(5.9)	(4.7)		(6.1)	(5.2)	
	Obs	83	82		80	92		67	107		71	101	
Experience: more than 16 years	Mean	39.8	36.6	3.2	45.0	38.0	7.0	41.8	27.1	14.7**	33.8	39.6	-5.8
	SE	(5.4)	(5.4)		(5.6)	(5.5)		(6.0)	(4.3)		(5.7)	(5.1)	
	Obs	83	82		80	92		67	107		71	101	
Wage: 3 to 4 minimum wages	Mean	30.1	19.5	10.6	31.3	22.8	8.4	10.4	27.1	-16.7***	19.7	21.8	-2.1
	SE	(5.1)	(4.4)		(5.2)	(4.5)		(3.8)	(4.8)		(4.7)	(4.2)	
	Obs	83	82		80	92		67	107		71	101	
Wage: 4 to 5 minimum wages	Mean	31.3	25.6	5.7	33.8	38.0	-4.3	32.8	20.6	12.3*	21.1	28.7	-7.6
	SE	(5.1)	(4.8)		(5.3)	(5.3)		(5.8)	(4.5)		(5.0)	(4.6)	
	Obs	83	82		80	92		67	107		71	101	
Wage: 5 to 6 minimum wages	Mean	4.8	17.1	-12.3**	16.3	16.3	-0.1	19.4	23.4	-4.0	25.4	29.7	-4.4
	SE	(2.4)	(4.2)		(4.2)	(4.0)		(4.8)	(4.2)		(5.1)	(4.7)	
	Obs	83	82		80	92		67	107		71	101	
Wage: more than 6 minimum wages	Mean	6.0	7.3	-1.3	6.3	9.8	-3.5	16.4	11.2	5.2	15.5	5.9	9.6**
	SE	(2.6)	(2.9)		(2.7)	(3.1)		(4.5)	(3.2)		(4.4)	(2.3)	
	Obs	83	82		80	92		67	107		71	101	

Notes: C = control, T = treatment; Diff is the difference between the two group means. Means and differences are in percentage points, with the standard error of each mean in parentheses. Group means and their standard errors come from a regression on a constant within each group; each difference is tested with a regression of the variable on treatment status and the strata dummies, with standard errors clustered at the school level; stars denote significance levels from that test: * 10%, ** 5%, *** 1%.
Source: Authors' estimates based on the pilot teacher questionnaire applied by CAEd at the end of the 2015 school year.

Table A.6: Pre-treatment balance test with administrative data at school level

	(1) Control		(2) Treatment		T-Test
	N	Mean/SE	N	Mean/SE	Difference (1)-(2)
% teachers in 1st to fifth grade with undergrad	84	96.18 (0.67)	86	96.67 (0.58)	-0.50
% teachers in 6th to ninth grade with undergrad	71	99.55 (0.21)	71	98.92 (0.29)	0.63*
Age grade distortion third grade (2013)	83	26.93 (1.89)	83	25.43 (1.85)	1.49***
Age grade distortion fifth grade (2013)	83	28.03 (1.65)	85	29.85 (2.01)	-1.82
Age grade distortion seventh grade (2013)	71	38.90 (1.97)	70	37.59 (2.14)	1.31*
Age grade distortion ninth grade (2013)	69	32.71 (2.18)	70	34.08 (2.71)	-1.37
Grade-promotion seventh grade (2009)	60	76.26 (1.96)	55	77.61 (1.73)	-1.35
Grade-promotion ninth grade (2009)	60	83.93 (1.63)	55	82.16 (1.97)	1.77**
Grade-promotion seventh grade (2011)	65	81.43 (1.22)	63	80.79 (1.37)	0.64
Grade-promotion ninth grade (2011)	64	87.69 (1.33)	63	85.81 (1.48)	1.88*
Grade-promotion seventh grade (2013)	69	84.39 (1.29)	68	83.48 (1.35)	0.91
Grade-promotion ninth grade (2013)	69	87.90 (1.20)	68	87.63 (1.25)	0.28
Grade-promotion third grade (2009)	68	77.66 (1.99)	74	78.54 (1.95)	-0.88
Grade-promotion fifth grade (2009)	68	83.16 (1.59)	74	82.83 (1.69)	0.33
Grade-promotion third grade (2011)	71	84.29 (1.37)	78	84.15 (1.40)	0.14
Grade-promotion fifth grade (2011)	71	88.93 (1.17)	79	87.77 (1.23)	1.16
Grade-promotion third grade (2013)	76	86.91 (1.02)	80	87.12 (1.08)	-0.21
Grade-promotion fifth grade (2013)	76	92.14 (0.85)	81	90.74 (1.05)	1.40
Math performance ninth grade (2009)	59	248.26 (4.00)	55	245.81 (3.59)	2.45
Reading performance ninth grade (2009)	59	244.89 (2.83)	55	245.24 (2.56)	-0.35
Math performance ninth grade (2011)	63	250.85 (4.09)	62	249.18 (3.87)	1.68
Reading performance ninth grade (2011)	63	246.88 (2.99)	62	242.68 (2.79)	4.20
Math performance ninth grade (2013)	69	252.60 (3.44)	67	249.20 (3.38)	3.39
Reading performance ninth grade (2013)	69	252.02 (2.56)	67	250.37 (2.46)	1.66
Math performance fifth grade (2009)	65	209.21 (4.24)	73	205.67 (3.66)	3.54
Reading performance fifth grade (2009)	65	189.33 (3.20)	73	186.64 (2.76)	2.69
Math performance fifth grade (2011)	69	216.10 (3.69)	79	211.92 (3.32)	4.18
Reading performance fifth grade (2011)	69	197.49 (2.87)	79	195.33 (2.56)	2.16
Math performance fifth grade (2013)	76	221.57 (3.83)	80	220.35 (3.73)	1.22
Reading performance fifth grade (2013)	76	206.25 (2.95)	80	204.90 (2.85)	1.36
IDEB ninth grade (2009)	59	3.88 (0.17)	55	3.87 (0.16)	0.01
IDEB ninth grade (2011)	63	4.15 (0.16)	62	4.04 (0.15)	0.11
IDEB ninth grade (2013)	69	4.33 (0.13)	67	4.24 (0.13)	0.08
IDEB fifth grade (2009)	65	4.60 (0.19)	73	4.49 (0.17)	0.12
IDEB fifth grade (2011)	69	4.98 (0.16)	79	4.86 (0.15)	0.12
IDEB fifth grade (2013)	76	5.40 (0.15)	80	5.32 (0.15)	0.08

Notes: Columns (1) and (2) report the control and treatment group means, with the standard error of each mean in parentheses; N is the number of schools. The difference column reports the control-minus-treatment difference between the two group means. Significance is from a regression of the covariate on treatment status and the six strata dummies, with standard errors clustered at the randomization-group level; stars denote significance levels: * 10%, ** 5%, *** 1%. Covariates are drawn from public administrative records (School Census, *Prova Brasil*, and IDEB) for the years shown.

Source: Authors' estimates based on public school-level administrative records from INEP (School Census, *Prova Brasil*, and IDEB).

Table A.7: Pre-treatment balance test for third and fifth graders

	(1) Control		(2) Treament		T-Test Difference
	N	Mean/SE	N	Mean/SE	(1)-(2)
Third grade					
Beneficiary of Bolsa Família	1082	36.23	1059	36.07	0.16
		(3.10)		(2.92)	
Adequate age for their grade	1038	86.99	997	85.66	1.34
		(1.49)		(1.76)	
Gender: male	2071	53.84	2009	51.57	2.27
		(0.92)		(1.22)	
Color: white	1624	38.92	1582	36.47	2.44
		(3.80)		(3.94)	
Mother education/Legal guardian: at least high school degree	1081	53.01	1050	50.95	2.05
		(2.59)		(2.26)	
Student found in School Census 2015-2018	2380	87.02	2270	88.50	-1.49
		(1.60)		(1.51)	
Student found in Prova Brasil 2017	2380	59.03	2270	59.16	-0.13
		(1.84)		(1.89)	
Prova Brasil reading score in 5th grade 2017	1405	232.76	1343	230.43	2.33
		(2.51)		(2.45)	
Prova Brasil math score in 5th grade 2017	1405	243.09	1343	238.55	4.54
		(2.92)		(3.09)	
Fifth grade					
Beneficiary of Bolsa Família	1905	45.35	1958	44.94	0.41
		(2.80)		(2.65)	
Adequate age for their grade	1902	78.39	1952	77.61	0.78
		(1.71)		(1.58)	
Gender: male	2302	51.87	2307	50.59	1.28
		(1.12)		(1.10)	
Color: white	1888	40.57	1945	37.33	3.25
		(2.52)		(2.48)	
Mother education/Legal guardian: at least high school degree	1272	55.66	1333	56.04	-0.38
		(1.87)		(2.00)	
Student found in School Census 2015-2018	2375	89.94	2380	90.04	-0.11
		(1.52)		(1.08)	
Student found in Prova Brasil 2015	2375	79.33	2380	81.60	-2.27
		(2.22)		(1.77)	
Prova Brasil reading score in 5th grade 2017	1884	220.58	1942	218.95	1.62
		(2.92)		(2.68)	
Prova Brasil math score in 5th grade 2017	1884	231.83	1942	229.17	2.66
		(3.82)		(3.33)	

Notes: We run the balance test controlling for strata. All variables are from *Prova Brasil* and the School Census. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Source: Authors' estimates based on the pilot roster merged with INEP's identified student-level administrative records (*Prova Brasil* and School Census); INEP performed the merge and the authors ran all balance tests inside INEP's secure facility.

Table A.8: Pre-treatment balance test for seventh and ninth graders

	(1) Control		(2) Treament		T-Test Difference
	N	Mean/SE	N	Mean/SE	(1)-(2)
Seventh grade					
Beneficiary of Bolsa Família	1790	40.95	1788	42.90	-1.95
		(2.92)		(2.87)	
Adequate age for their grade	1781	77.20	1781	76.81	0.39
		(2.18)		(1.85)	
Gender: male	1798	48.00	1799	48.08	-0.08
		(1.45)		(1.04)	
Color: white	2226	31.63	2186	32.20	-0.58
		(4.09)		(3.89)	
Mother education/Legal guardian: at least high school degree	1347	52.64	1317	54.75	-2.11
		(2.01)		(2.05)	
Student found in School Census 2015-2018	2376	91.20	2321	91.12	0.08
		(1.66)		(1.11)	
Student found in Prova Brasil 2013	2376	67.05	2321	66.52	0.52
		(2.25)		(1.91)	
Prova Brasil reading score in 5th grade 2013	1593	211.03	1544	209.28	1.75
		(3.16)		(2.68)	
Prova Brasil math score in 5th grade 2013	1593	226.11	1544	224.70	1.41
		(4.10)		(3.66)	
Student found in Prova Brasil 2017	2376	62.50	2321	61.83	0.67
		(2.04)		(1.88)	
Prova Brasil reading score in 9th grade 2017	1485	273.81	1435	274.74	-0.93
		(2.58)		(2.31)	
Prova Brasil math score in 9th grade 2017	1485	269.66	1435	271.07	-1.42
		(3.48)		(3.01)	
Ninth grade					
Beneficiary of Bolsa Família	1648	38.35	1604	35.79	2.56*
		(3.07)		(2.59)	
Adequate age for their grade	1636	76.96	1602	76.09	0.86
		(1.80)		(2.30)	
Gender: male	2161	49.56	2174	47.10	2.46*
		(1.05)		(1.08)	
Color: white	2074	32.64	2087	30.09	2.55
		(4.22)		(3.95)	
Mother education/Legal guardian: at least high school degree	1364	57.26	1335	57.00	0.25
		(1.75)		(1.77)	
Student found in School Census 2015-2018	2222	92.26	2233	92.79	-0.53
		(0.98)		(0.84)	
Student found in Prova Brasil 2011	2222	63.77	2233	64.22	-0.45
	(72)	(2.24)	(72)	(2.06)	
Prova Brasil reading score in 5th grade 2011	1417	206.10	1434	200.39	5.71**
	(71)	(3.07)	(72)	(2.78)	
Prova Brasil math score in 5th grade 2011	1417	221.42	1434	215.72	5.69
	(71)	(3.80)	(72)	(3.59)	
Student found in Prova Brasil 2015	2222	82.36	2233	81.10	1.26
	(72)	(1.31)	(72)	(1.45)	
Prova Brasil reading score in 9th grade 2015	1830	265.01	1811	264.14	0.87
	(70)	(2.27)	(70)	(1.92)	
Prova Brasil math score in 9th grade 2015	1830	264.88	1811	260.73	4.15*
	(70)	(2.73)	(70)	(2.74)	
Graduated from high school	2222	59.63	2233	56.02	3.61
	(72)	(1.58)	(72)	(1.91)	
Participated in ENEM, 2018	2222	46.17	2233	43.62	2.56
	(72)	(1.54)	(72)	(1.83)	
Standardized ENEM score, 2018	793	0.13	723	0.12	0.01
	(71)	(0.04)	(71)	(0.05)	

Notes: We run the balance test controlling for strata. All variables are from *Prova Brasil* and the School Census. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Source: Authors' estimates based on the pilot roster merged with INEP's identified student-level administrative records (*Prova Brasil* and School Census); INEP performed the merge and the authors ran all balance tests inside INEP's secure facility.

Table A.9: Average causal mediation effects (mediator: financial proficiency)

	Consumption		Saving		Talks with parents		Talks with friends		Piggy bank use		Use of financial services		Receiving an allowance	
	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI
Pooled														
Total effect	0.091	[0.053, 0.128]	0.048	[0.008, 0.086]	0.015	[-0.004, 0.034]	0.036	[0.019, 0.053]	0.016	[-0.003, 0.035]	0.007	[-0.010, 0.023]	0.017	[0.000, 0.033]
ADE	0.059	[0.024, 0.094]	0.026	[-0.012, 0.064]	0.013	[-0.006, 0.032]	0.038	[0.020, 0.055]	0.015	[-0.005, 0.034]	0.010	[-0.007, 0.026]	0.017	[0.001, 0.034]
ACME	0.032	[0.016, 0.049]	0.022	[0.010, 0.034]	0.002	[0.001, 0.003]	-0.001	[-0.002, -0.000]	0.001	[0.001, 0.003]	-0.003	[-0.005, -0.001]	-0.000	[-0.001, 0.000]
% mediated	0.347	[0.248, 0.601]	0.440	[0.239, 1.717]	0.096	[-0.869, 0.781]	-0.033	[-0.064, -0.023]	0.080	[-0.363, 0.911]	-0.274	[-4.343, 3.196]	-0.020	[-0.114, -0.009]
Primary education students														
Total effect	0.128	[0.061, 0.193]	0.085	[0.017, 0.154]	0.059	[0.027, 0.092]	0.056	[0.030, 0.082]	0.019	[-0.011, 0.050]	0.002	[-0.025, 0.029]	0.016	[-0.011, 0.042]
ADE	0.107	[0.047, 0.170]	0.071	[0.006, 0.138]	0.060	[0.028, 0.092]	0.058	[0.032, 0.085]	0.018	[-0.012, 0.049]	0.005	[-0.022, 0.033]	0.016	[-0.011, 0.043]
ACME	0.020	[-0.003, 0.043]	0.014	[-0.006, 0.033]	-0.000	[-0.001, 0.001]	-0.002	[-0.005, 0.001]	0.001	[-0.000, 0.003]	-0.003	[-0.007, 0.001]	-0.000	[-0.001, 0.001]
% mediated	0.159	[0.105, 0.333]	0.162	[0.089, 0.701]	-0.004	[-0.010, -0.003]	-0.041	[-0.076, -0.027]	0.051	[-0.378, 0.590]	-0.125	[-4.469, 4.256]	-0.000	[-0.002, 0.001]
Middle school students														
Total effect	0.073	[0.027, 0.119]	0.029	[-0.017, 0.078]	-0.013	[-0.036, 0.011]	0.025	[0.004, 0.048]	0.009	[-0.015, 0.035]	0.008	[-0.011, 0.029]	0.020	[-0.000, 0.041]
ADE	0.035	[-0.006, 0.079]	0.004	[-0.041, 0.052]	-0.016	[-0.039, 0.008]	0.025	[0.004, 0.048]	0.007	[-0.016, 0.033]	0.011	[-0.009, 0.032]	0.020	[0.000, 0.042]
ACME	0.038	[0.017, 0.059]	0.025	[0.011, 0.039]	0.003	[0.001, 0.005]	0.000	[-0.001, 0.001]	0.002	[0.001, 0.004]	-0.002	[-0.004, -0.001]	-0.001	[-0.002, 0.000]
% mediated	0.520	[0.320, 1.356]	0.675	[-11.717, 8.456]	-0.185	[-3.029, 1.834]	0.014	[0.007, 0.075]	0.107	[-2.784, 1.571]	-0.164	[-3.053, 2.998]	-0.028	[-0.213, 0.105]

Notes: Each dependent variable equals 1 if the student talks to their parents or friends about financial subjects, keeps a piggy bank, has access to financial services, or receives an allowance, and 0 otherwise; consumption and savings are the standardized attitude indices. The mediator is the financial-proficiency index. *Total effect* is the total treatment effect (τ), *ADE* the average direct effect, *ACME* the average causal mediation effect operating through financial proficiency, and *% mediated* the share of the total effect that runs through the mediator. Estimated with the `medeff` command; 95% confidence intervals in brackets from 1,000 simulations. Controls: strata, school infrastructure, and syllabus complexity. The sample is restricted to fifth-, seventh-, and ninth-grade students: third graders, who answered only the piggy-bank and financial-services items, are excluded so that all outcomes cover the same grades.
Source: Authors' estimates based on the pilot questionnaires applied by CAEd to students at the end of the 2015 school year, merged with school-level administrative records.

Table A.10: ITT estimates controlling for pre-pilot performance

	Seventh grade		Ninth grade	
	(1) Strata FE	(2) Strata FE + pre-pilot performance	(3) Strata FE	(4) Strata FE + pre-pilot performance
Financial proficiency				
Treatment	0.090*	0.098**	0.107*	0.149***
	(0.048)	(0.042)	(0.056)	(0.055)
RI pvalue	0.061	0.022	0.057	0.008
N. obs	3640	2538	3374	2251
R-squared	0.208	0.505	0.168	0.399
Consumption index				
Treatment	0.080*	0.070*	0.084**	0.087*
	(0.041)	(0.042)	(0.037)	(0.044)
RI pvalue	0.054	0.098	0.025	0.050
N. obs	3409	2384	3163	2115
R-squared	0.100	0.187	0.055	0.128
Saving index				
Treatment	0.066	0.033	0.018	0.030
	(0.040)	(0.044)	(0.038)	(0.045)
RI pvalue	0.102	0.456	0.644	0.508
N. obs	3368	2351	3105	2080
R-squared	0.032	0.065	0.019	0.044

Notes: Clustered standard errors at the school level in parentheses; RI denotes randomization inference. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Columns (2) and (4) add the student's fifth-grade *Prova Brasil* reading and math scores, taken in 2013 for the seventh-grade cohort and in 2011 for the ninth-grade cohort, as controls; the smaller sample in these columns reflects the students the merge with INEP's records located in those waves. Financial proficiency is computed by CAEd from students' answers to a standardized test scored with Item Response Theory. The consumption and savings indices summarize students' answers to fifteen agreement-scale statements, reported with their weights in Table A.3. All three indices are normalized to standard-deviation units, so treatment effects are read in standard deviations.
Source: Authors' estimates based on the pilot roster merged with INEP's identified student-level administrative records (School Census and *Prova Brasil*). INEP performed the merge; the authors ran all estimates inside INEP's secure facility.

Table A.11: First-stage and local average treatment effect (LATE) estimates

	Pooled		Elementary		Middle school	
	LATE (i)	LATE (ii)	LATE (i)	LATE (ii)	LATE (i)	LATE (ii)
Financial proficiency						
Treatment	0.083**	0.098**	0.055	0.063	0.112***	0.135**
	(0.039)	(0.046)	(0.053)	(0.061)	(0.048)	(0.058)
pvalue RW	0.023	0.023	0.165	0.165	0.009	0.012
First stage: assignment	0.870	0.742	0.861	0.754	0.878	0.729
	(0.022)	(0.019)	(0.029)	(0.026)	(0.032)	(0.026)
First-stage F	1601.0	1541.2	887.5	837.1	741.6	786.5
N. obs	14655	14655	7641	7641	7014	7014
Consumption index						
Treatment	0.109***	0.136***	0.150***	0.202***	0.093***	0.112***
	(0.031)	(0.039)	(0.052)	(0.070)	(0.035)	(0.043)
pvalue RW	0.001	0.001	0.003	0.003	0.006	0.006
First stage: assignment	0.862	0.689	0.815	0.605	0.886	0.734
	(0.025)	(0.021)	(0.043)	(0.036)	(0.032)	(0.025)
First-stage F	1170.4	1065.1	359.3	281.8	786.0	881.7
N. obs	10220	10220	3648	3648	6572	6572
Saving index						
Treatment	0.063**	0.079**	0.107**	0.143**	0.050	0.060
	(0.029)	(0.037)	(0.051)	(0.069)	(0.033)	(0.040)
pvalue RW	0.023	0.023	0.023	0.023	0.103	0.103
First stage: assignment	0.861	0.690	0.814	0.606	0.885	0.734
	(0.025)	(0.021)	(0.043)	(0.037)	(0.032)	(0.025)
First-stage F	1160.7	1031.6	357.4	275.0	767.8	847.0
N. obs	10022	10022	3549	3549	6473	6473

Notes: Clustered standard errors at the school level in parentheses. Each column reproduces one of the LATE estimates of Table 6 and reports, below it, its own first stage: the coefficient of random assignment on exposure from Equation 2, estimated by ordinary least squares on the same sample as the second stage, with its clustered standard error in parentheses and the F statistic of the excluded instrument. Because each outcome is observed for a slightly different set of students, the estimation sample changes across panels, and the first-stage coefficient of the same column therefore varies slightly from one outcome to another. RW denotes Romano-Wolf stepdown p-values (1,000 bootstrap replications resampling school clusters within randomization strata), from the same 27-regression family of Table 6, and stars denote significance levels based on the Romano-Wolf adjusted p-value: * 10%, ** 5%, *** 1%. Exposure is measured under the conservative (i) and broad (ii) definitions described in the text. All three indices are normalized to standard-deviation units. The consumption and savings indices are unavailable for third graders, who did not answer these items. Therefore, elementary estimates represent the fifth grade, while middle school estimates represent the seventh and ninth grades.
Source: Authors' estimates based on the pilot questionnaires applied by CAEd to students at the end of the 2015 school year.